

# ECE/CS 757 Advanced Computer Architecture II

## Spring 2025

### Cache and Memory Hierarchy

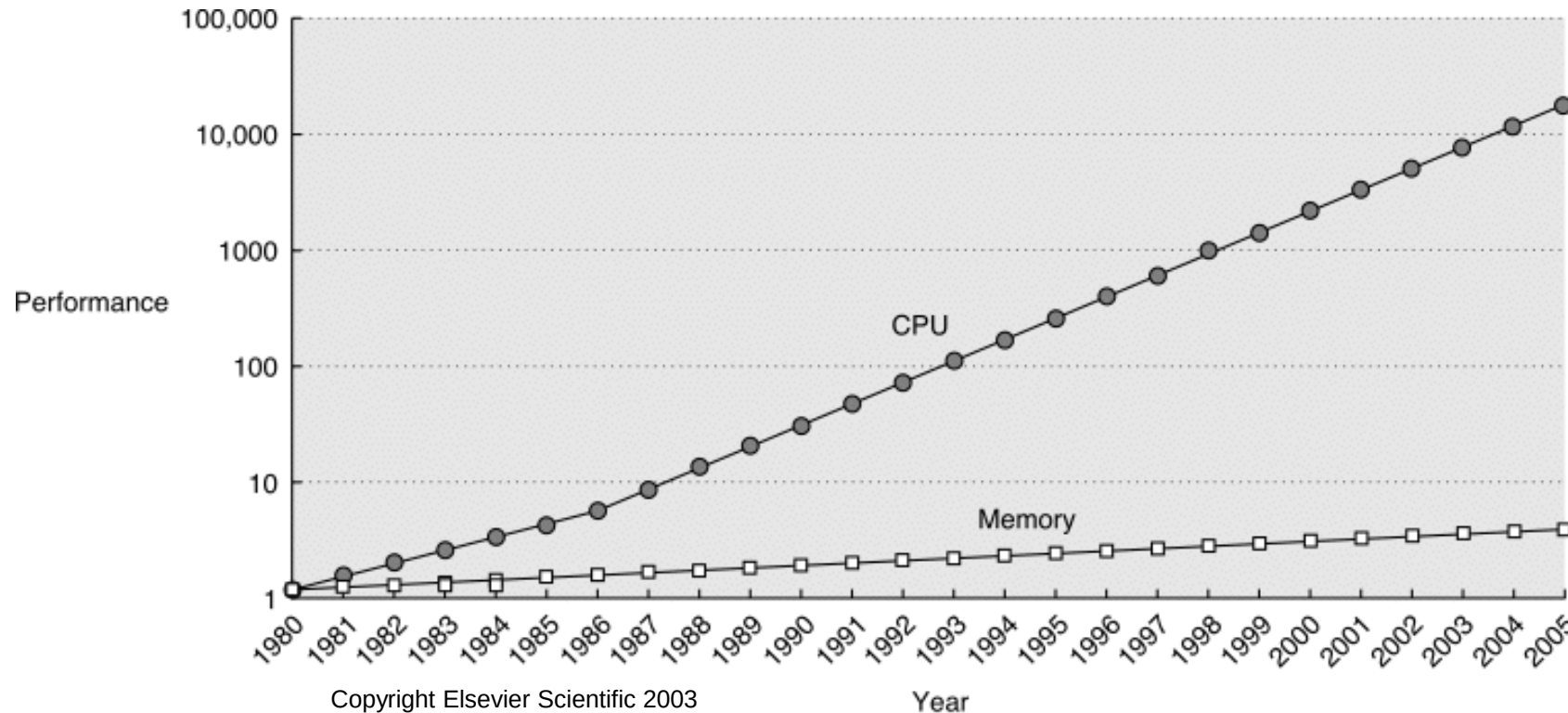
**Instructor:** Prof. Joshua San Miguel

Some slides adopted from Mikko Lipasti

# Memory Wall

Processors have been getting faster more quickly than memory

- Processor speed improvement: 35%-55%
- Memory latency improvement: 7%

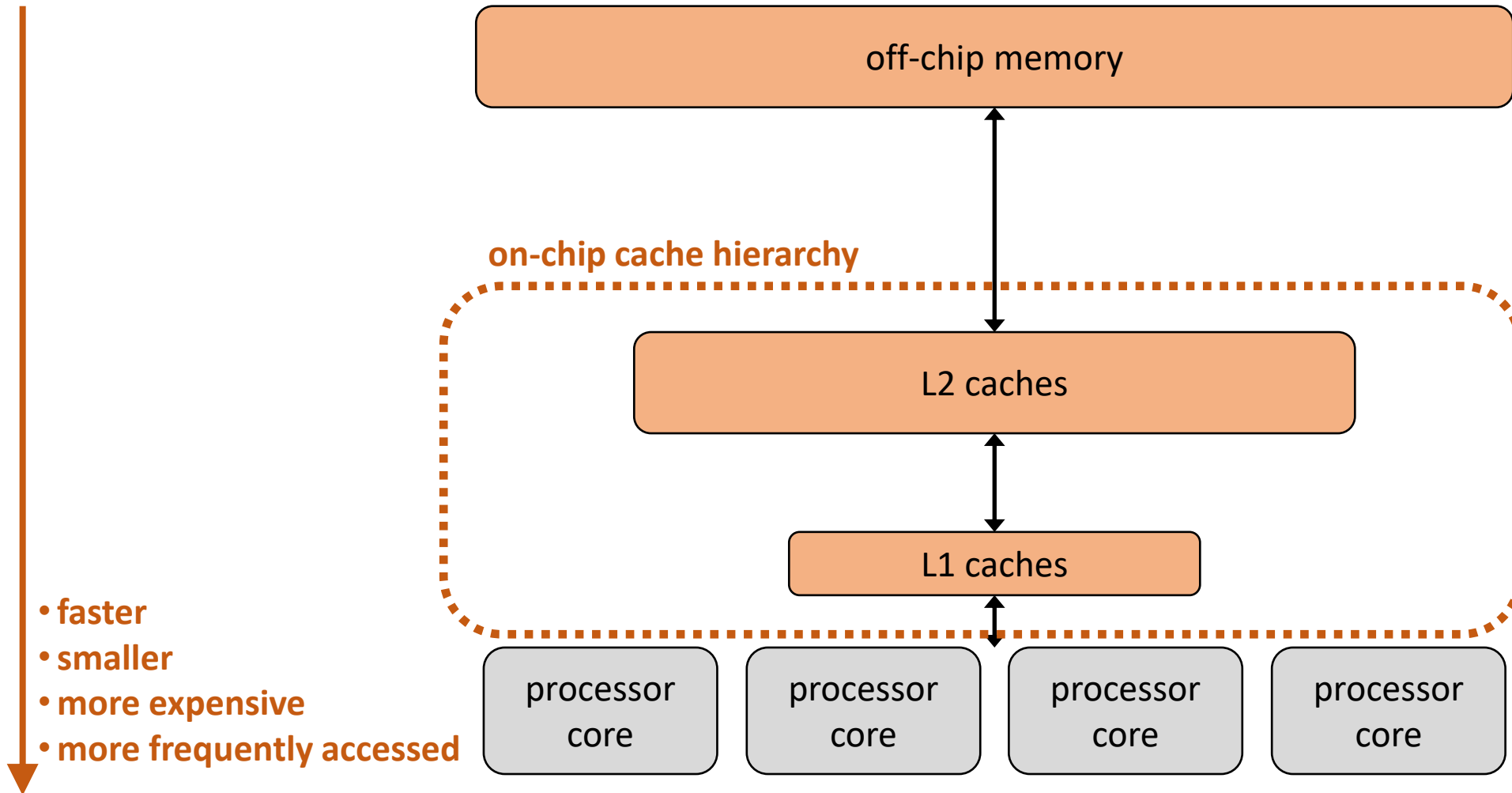


# Memory Locality

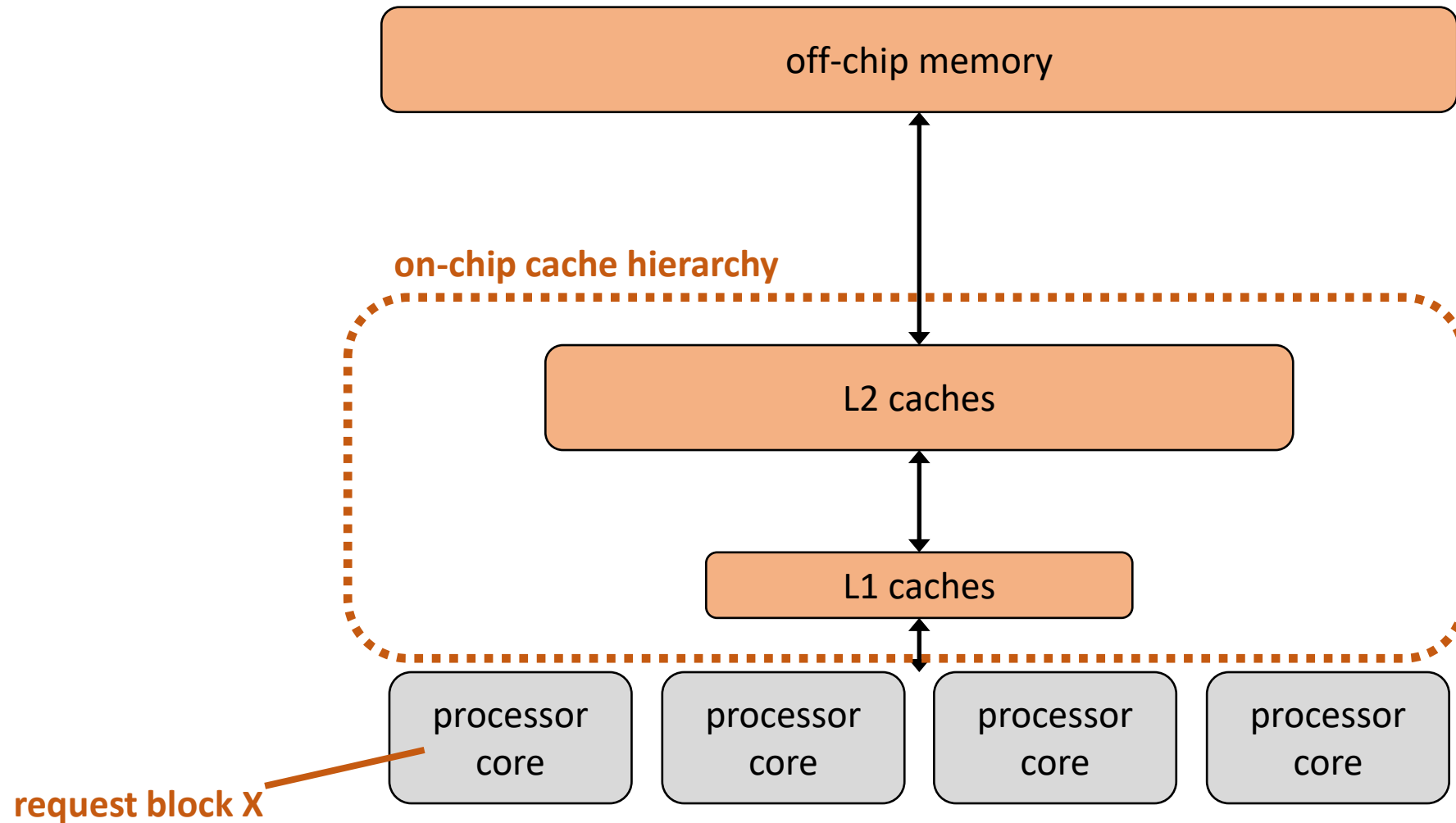
## Memory locality to the rescue

- Property of real programs
- Holds for data and instructions
- **Temporal locality:**
  - Recently-referenced data is likely to be referenced again soon
  - **Reactive:** cache recently used data in small, fast memory
- **Spatial locality:**
  - More likely to reference data near recently-referenced data
  - **Proactive:** fetch data in large chunks to include nearby data

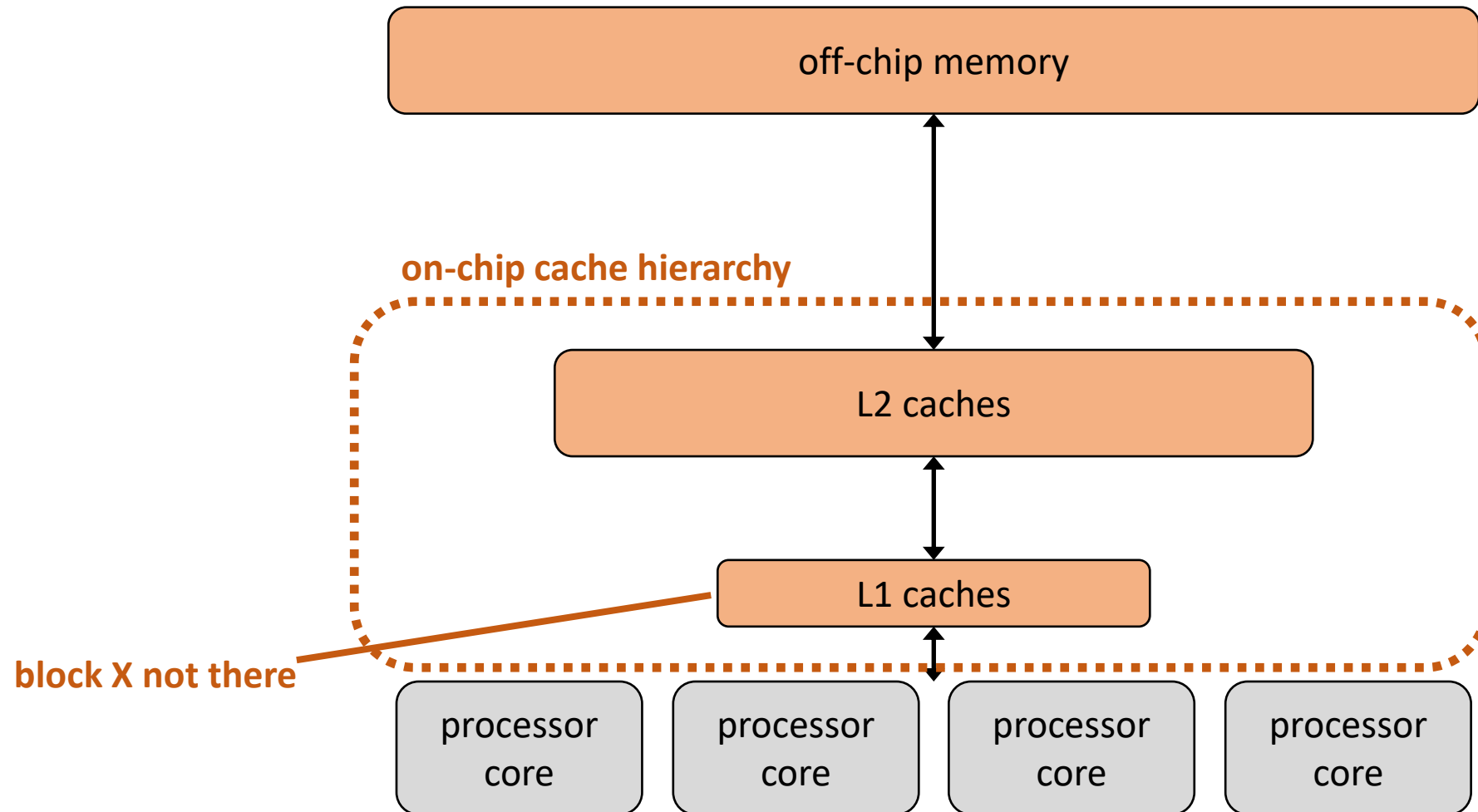
# Shared-Memory CMP



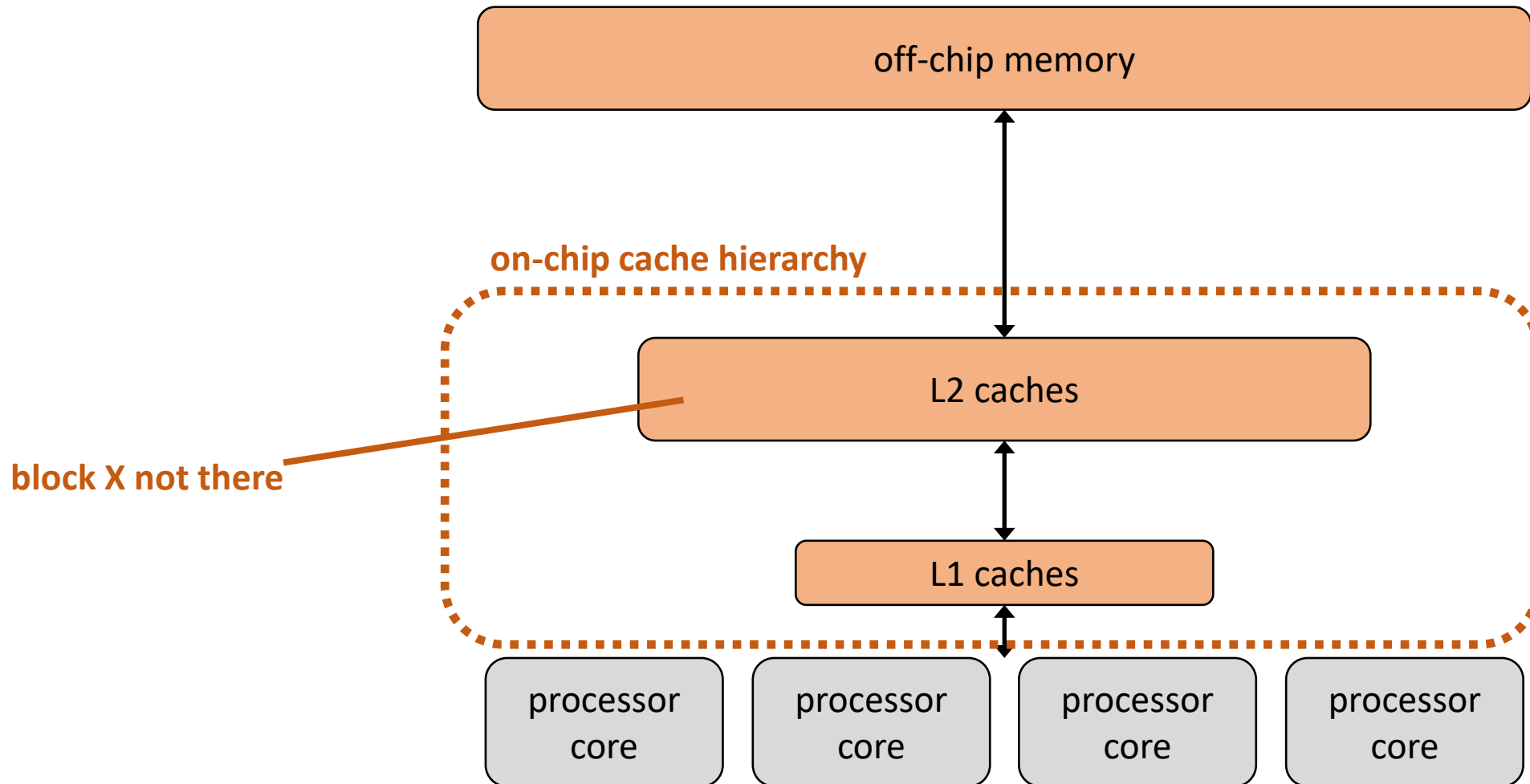
# Shared-Memory CMP – Memory Access



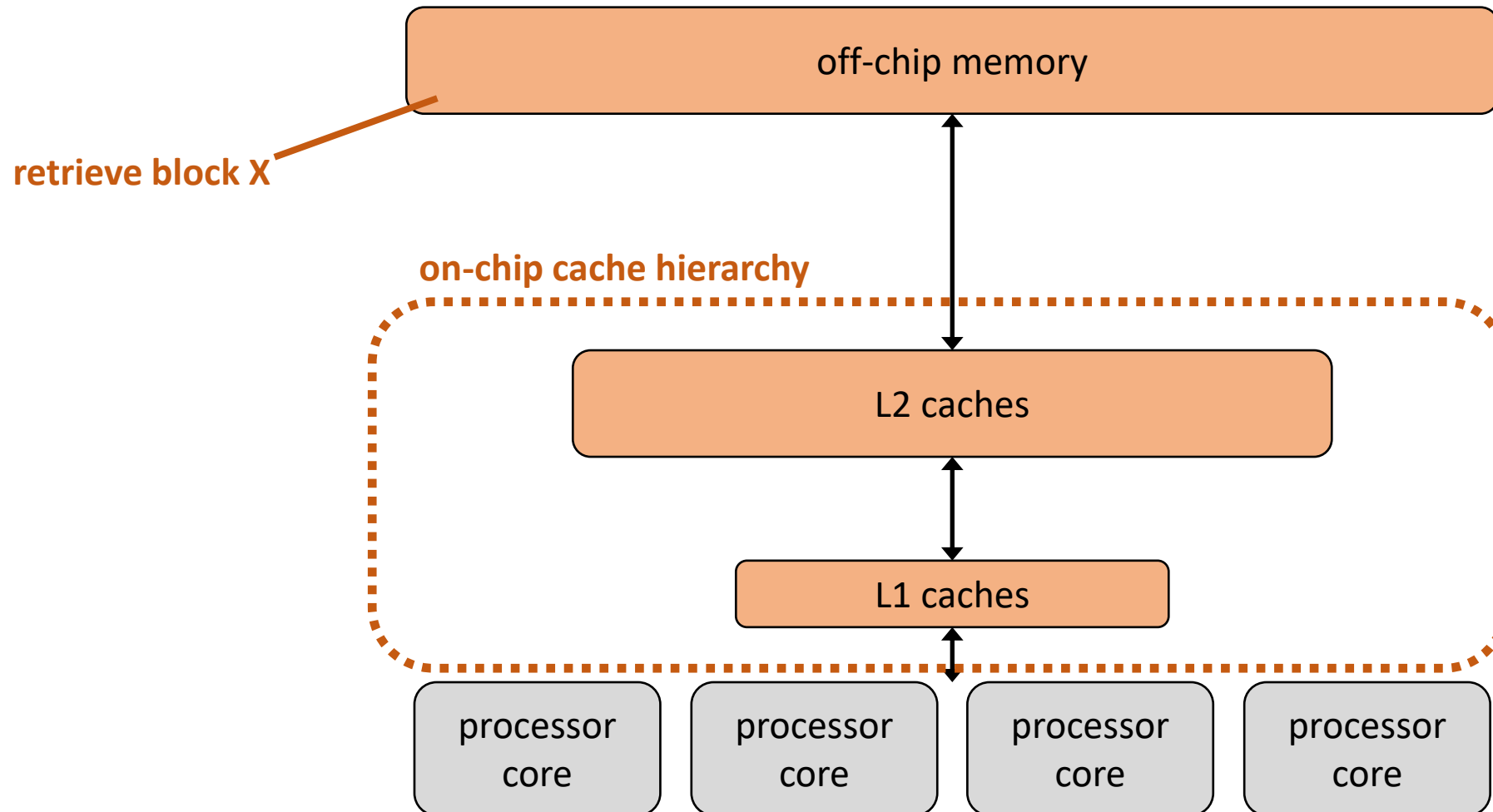
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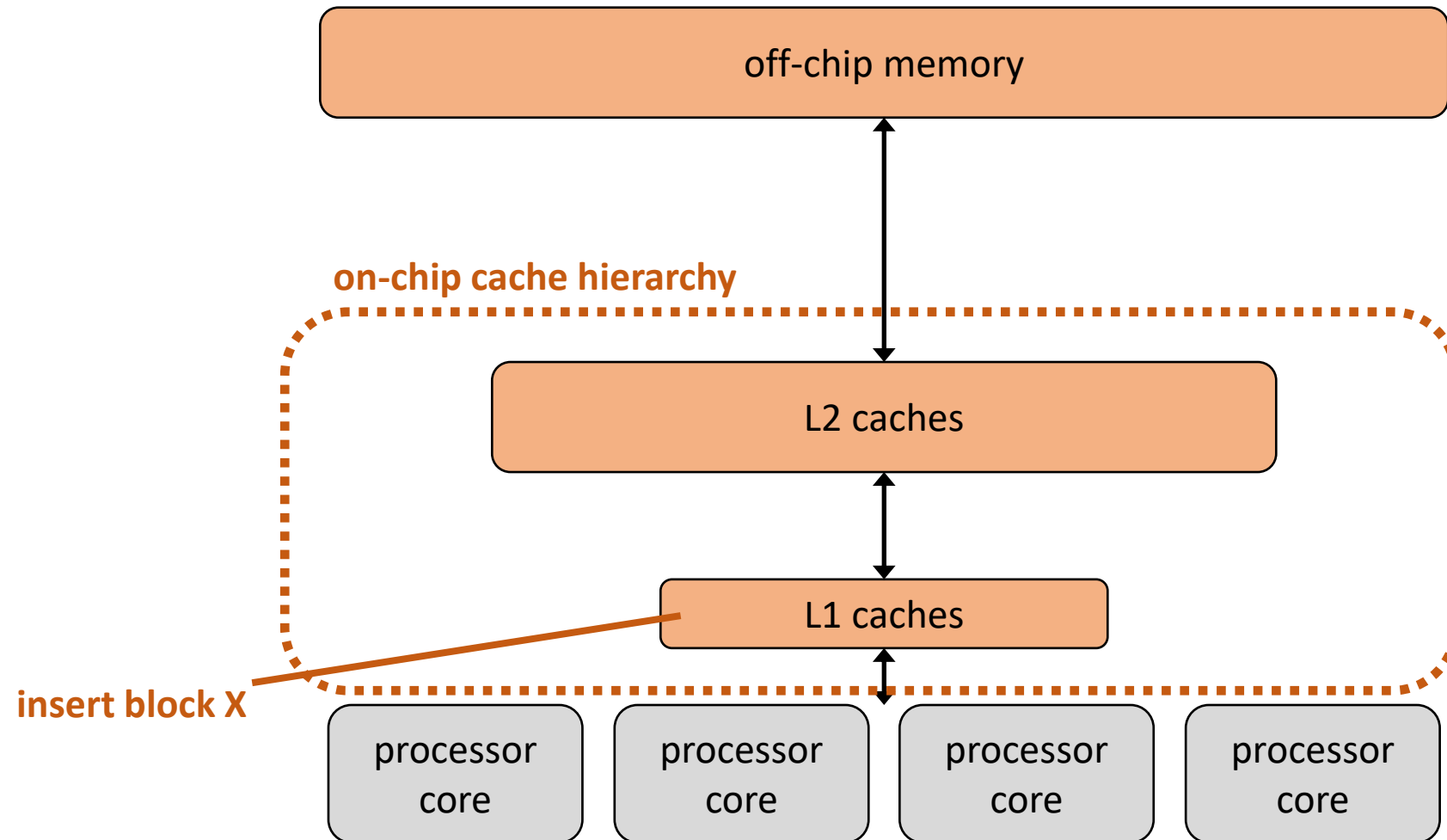
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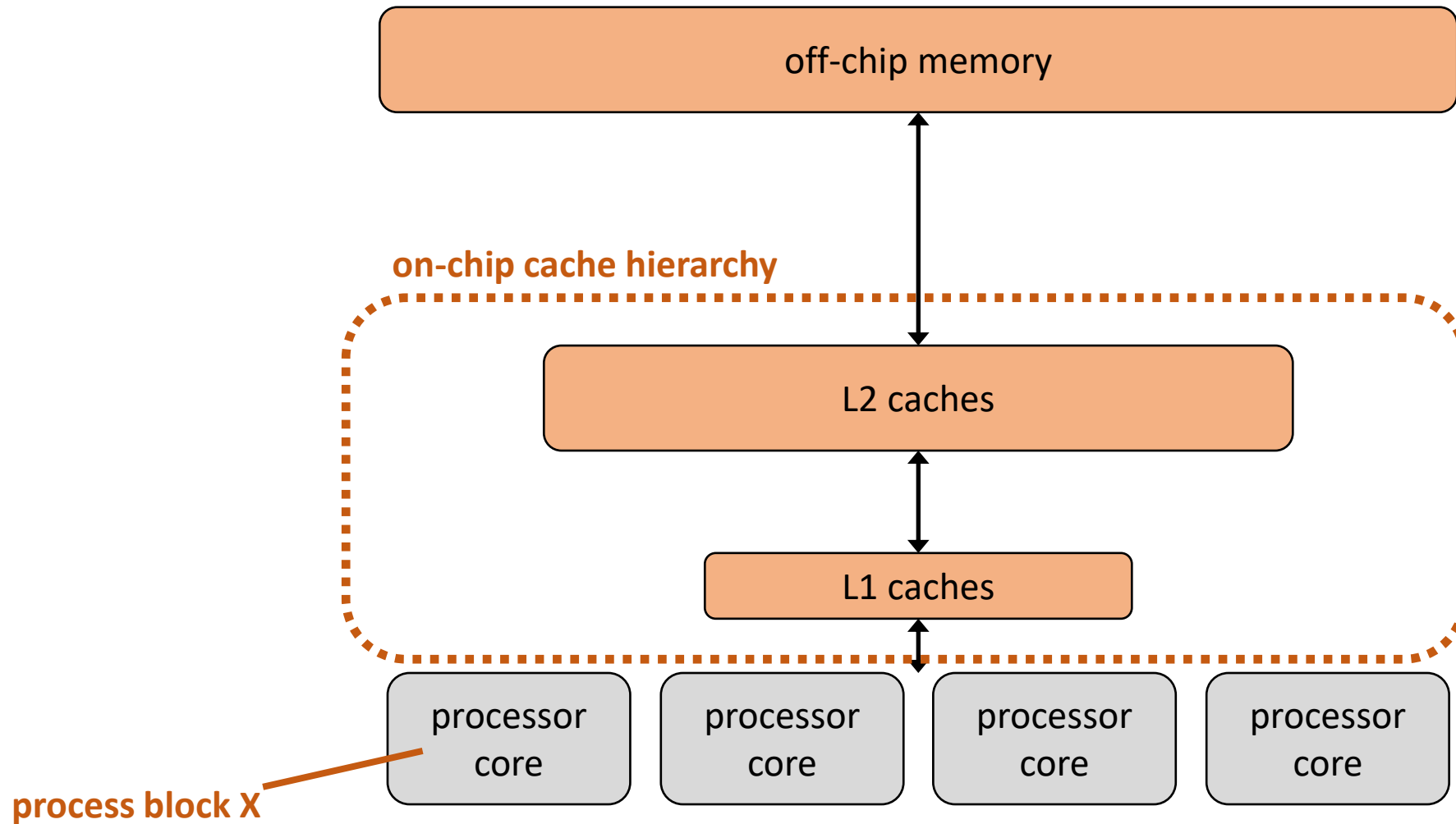
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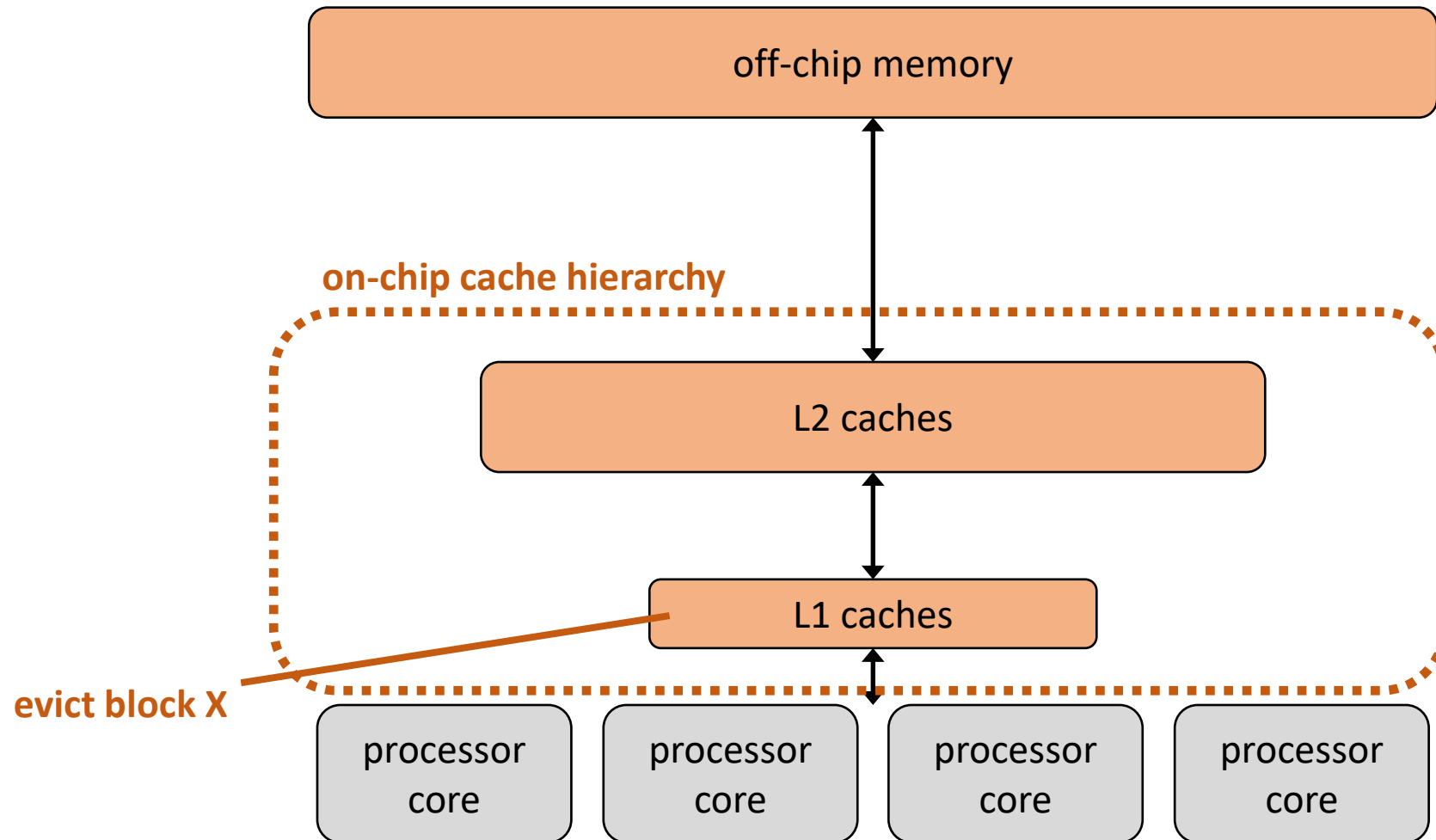
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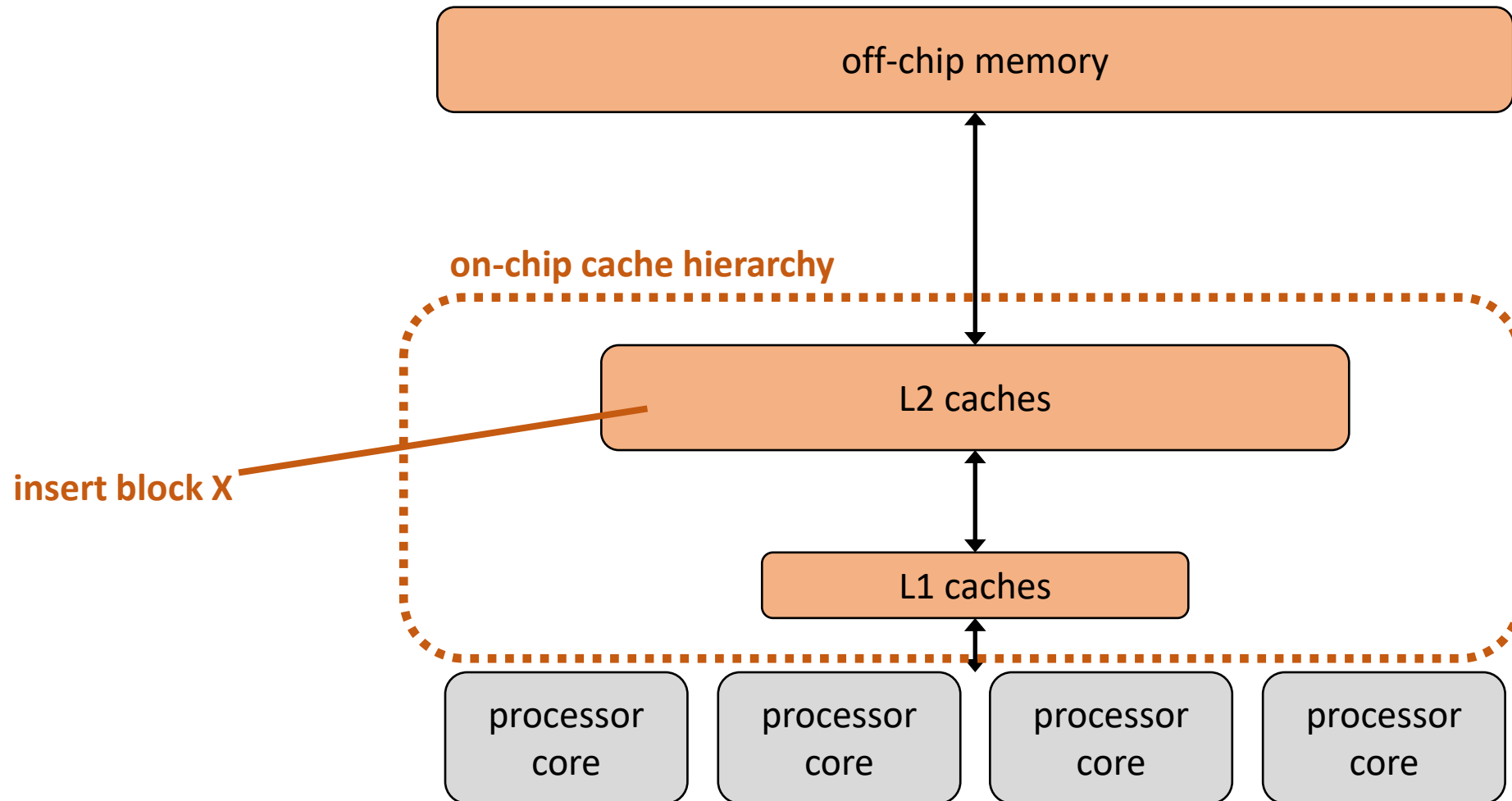
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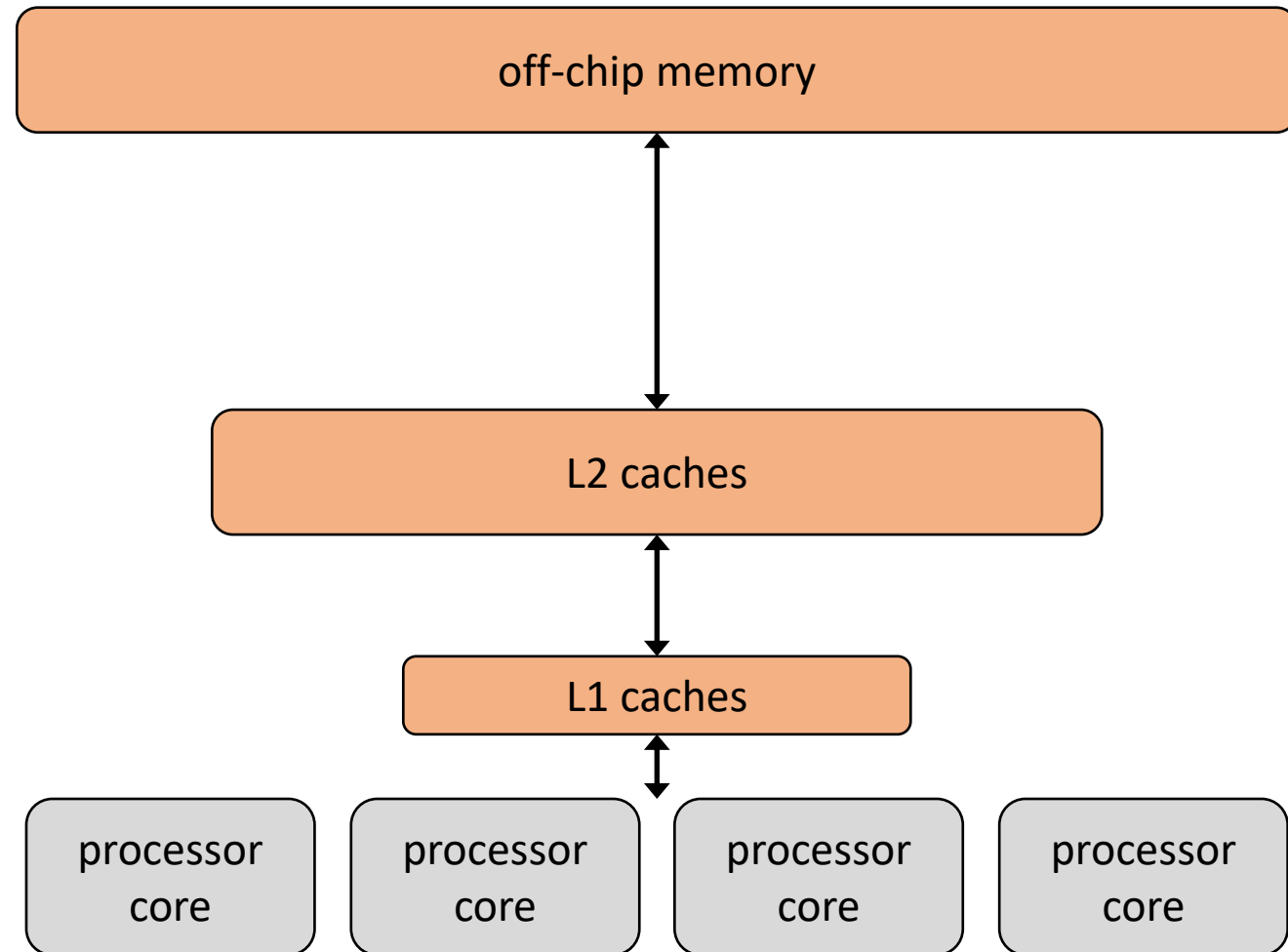
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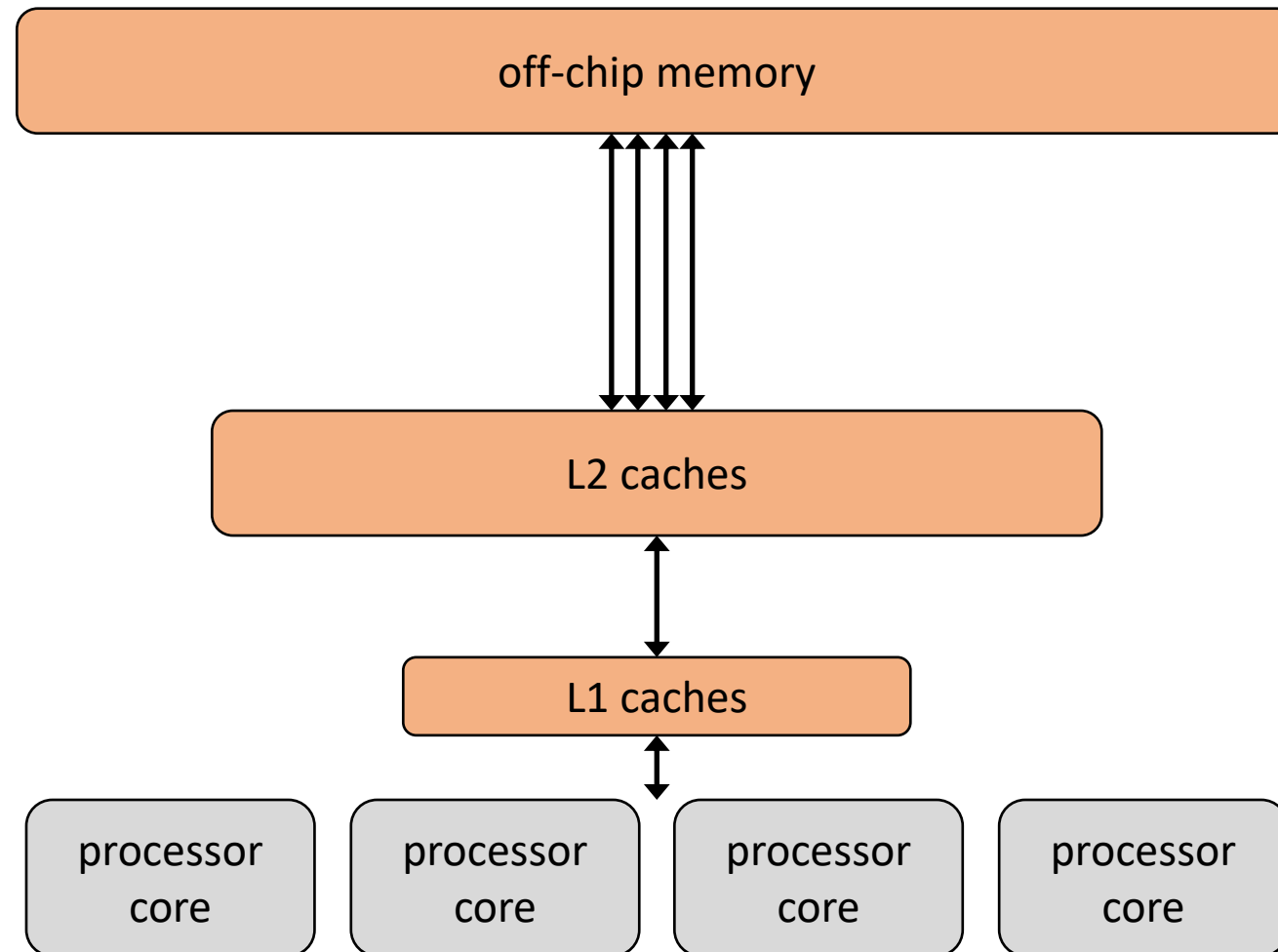
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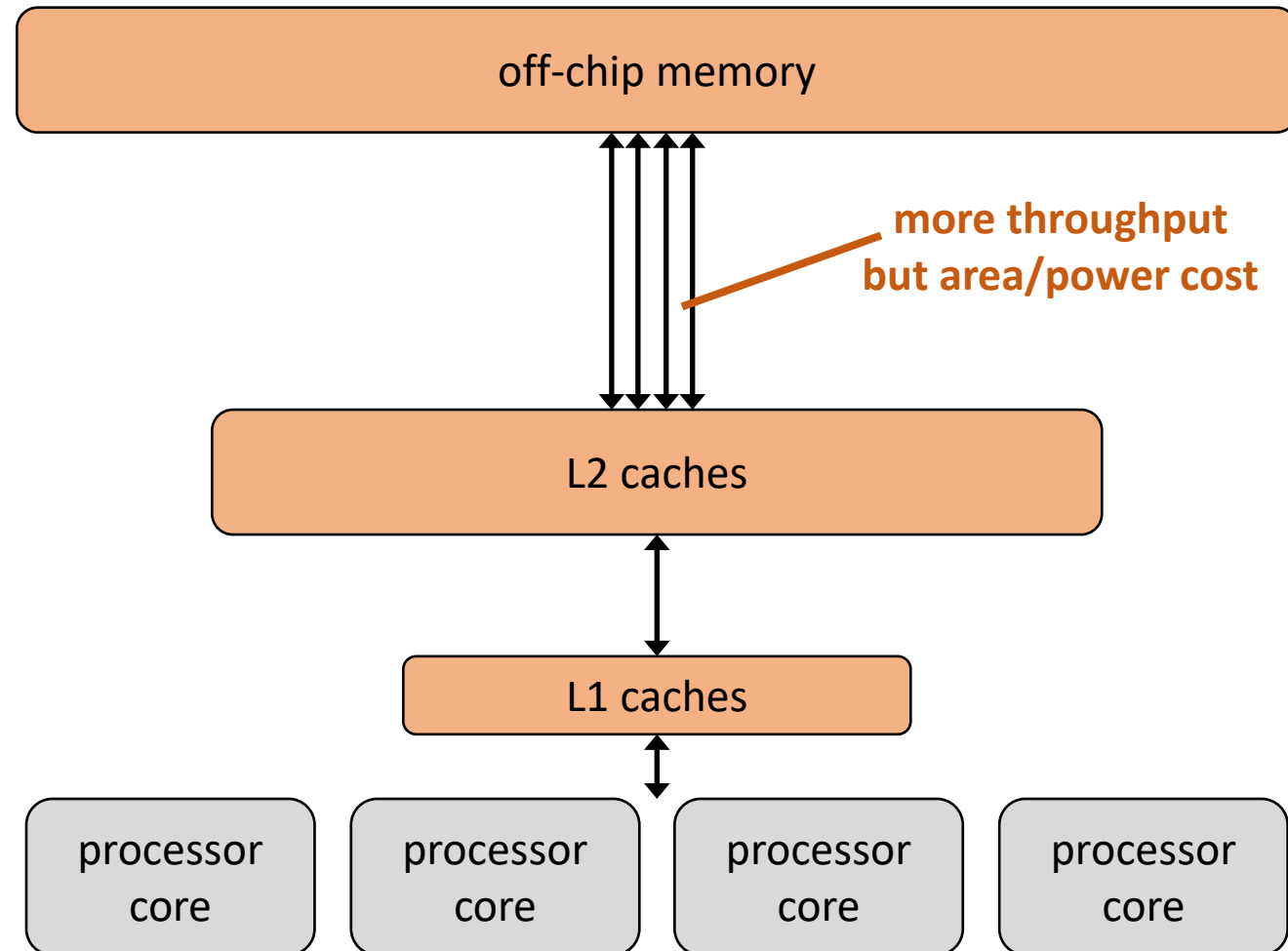
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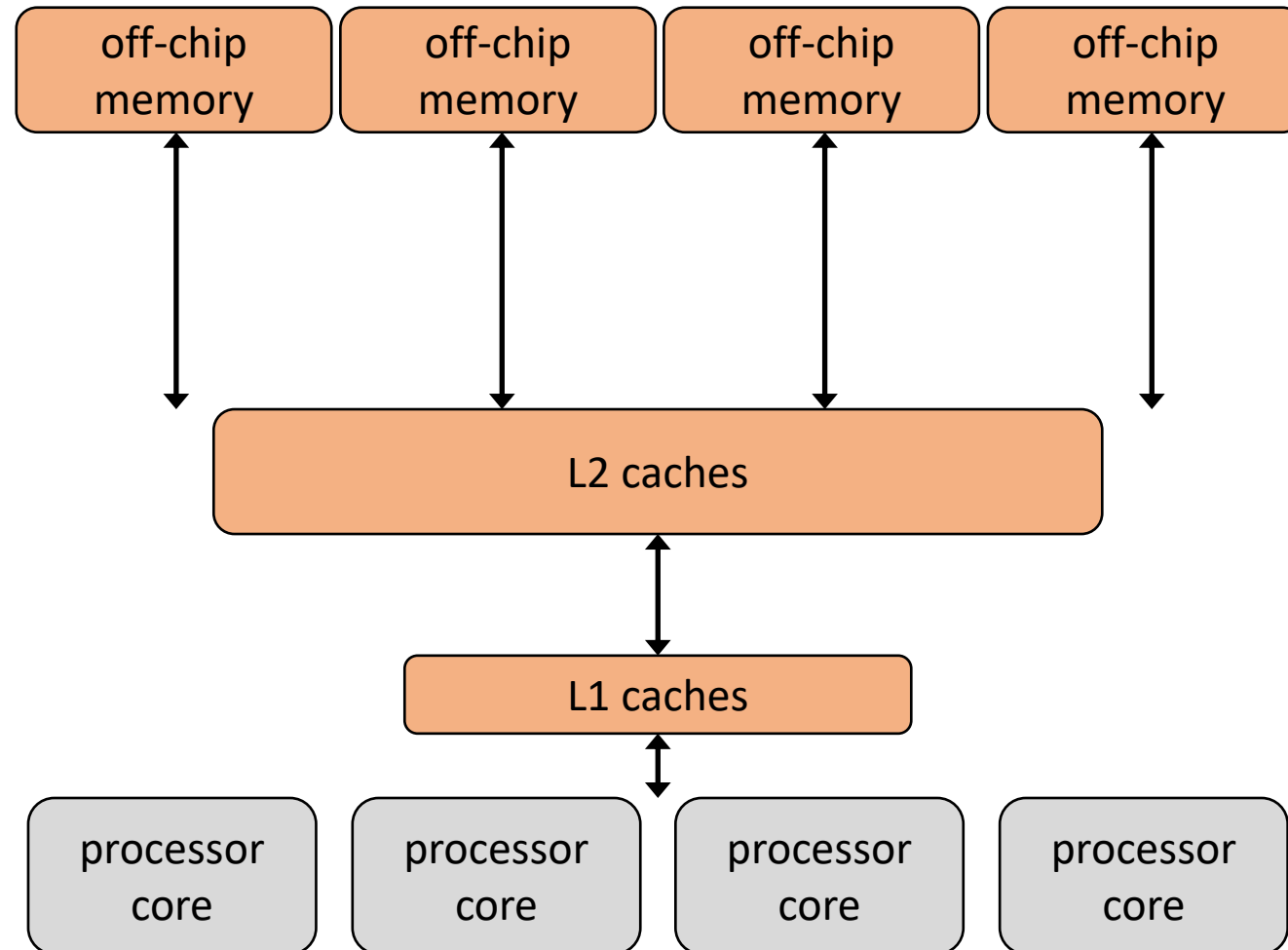
# Shared-Memory CMP – Wider Memory



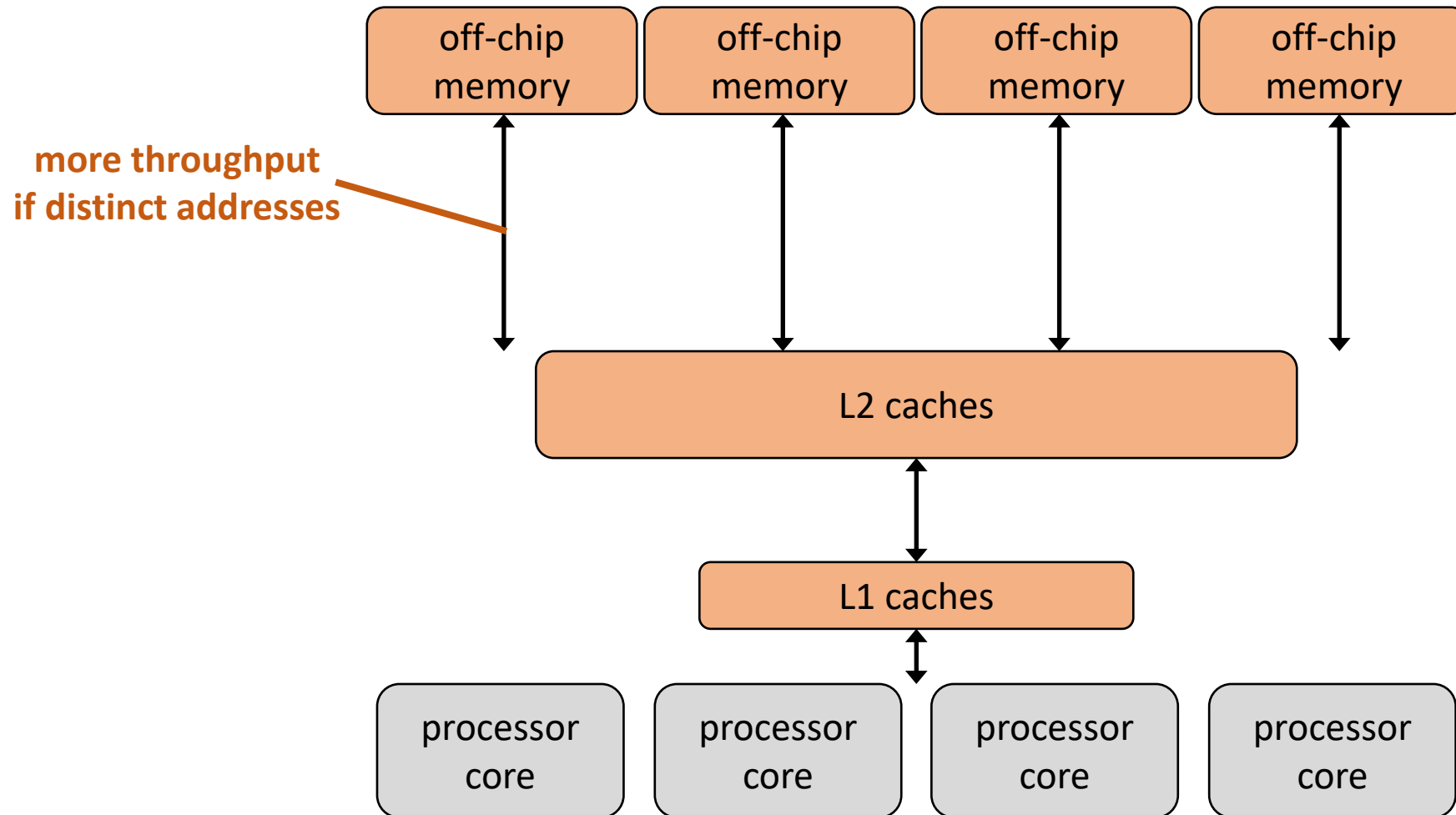
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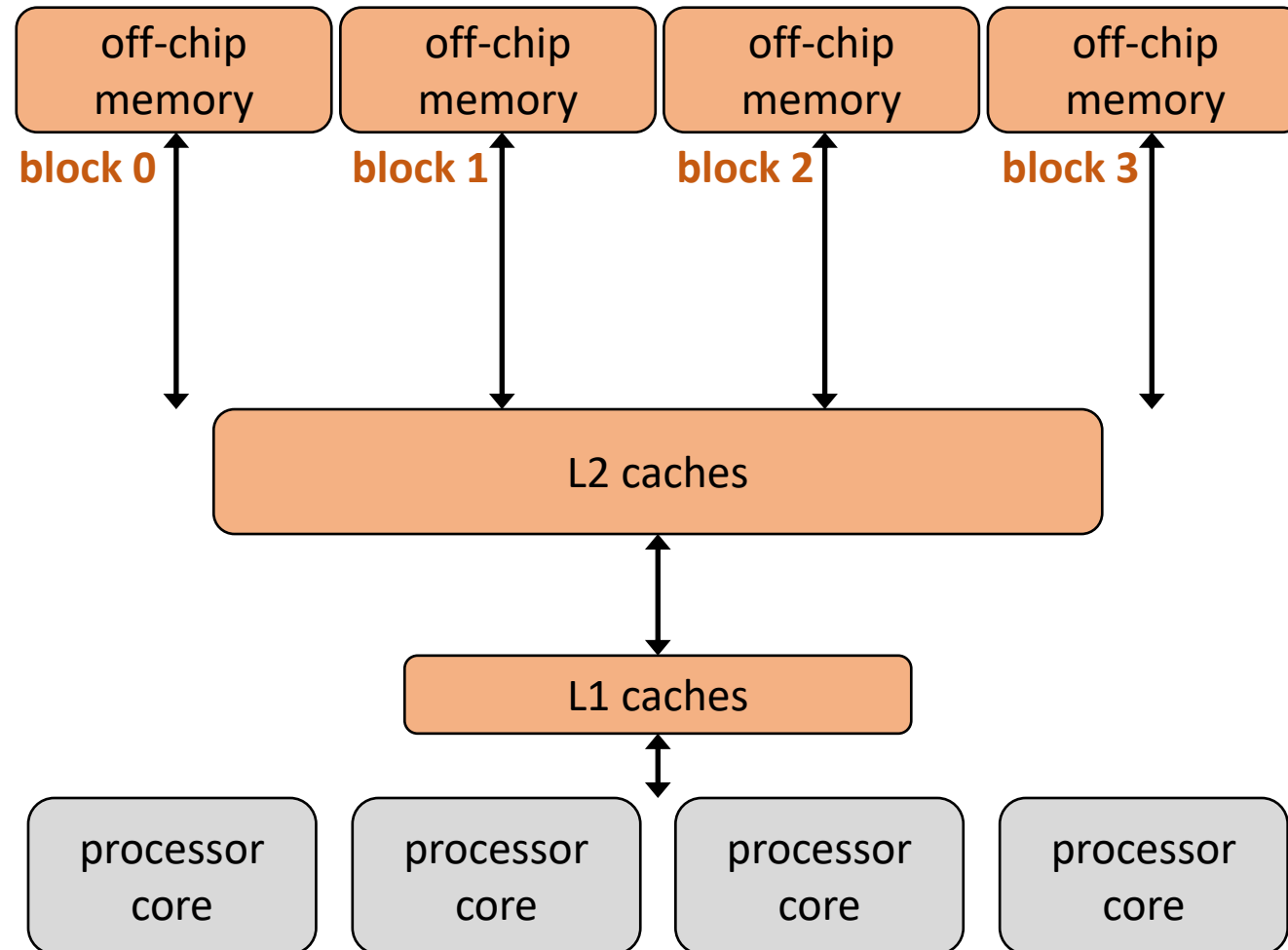
# Shared-Memory CMP – Banked Memory



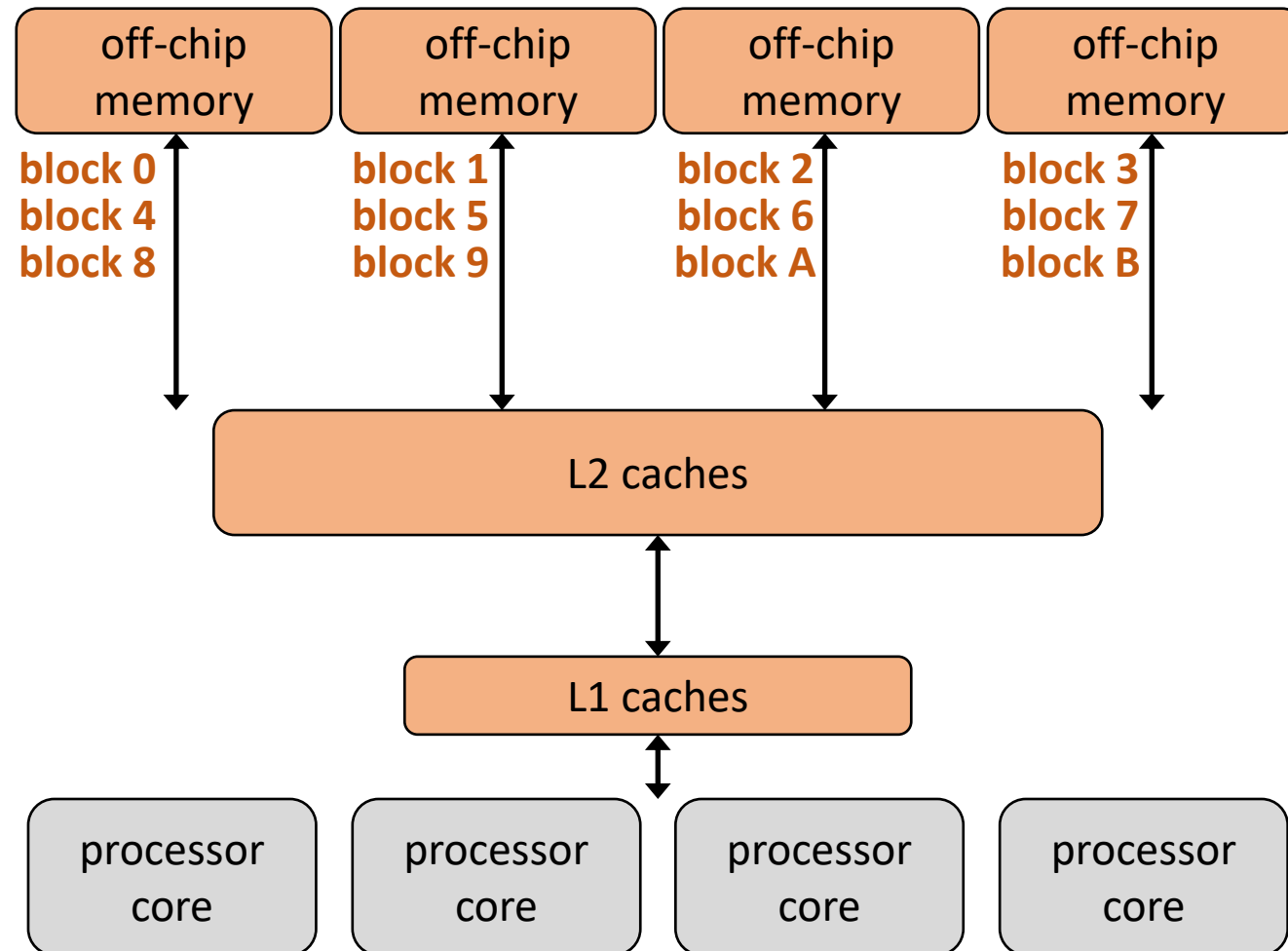
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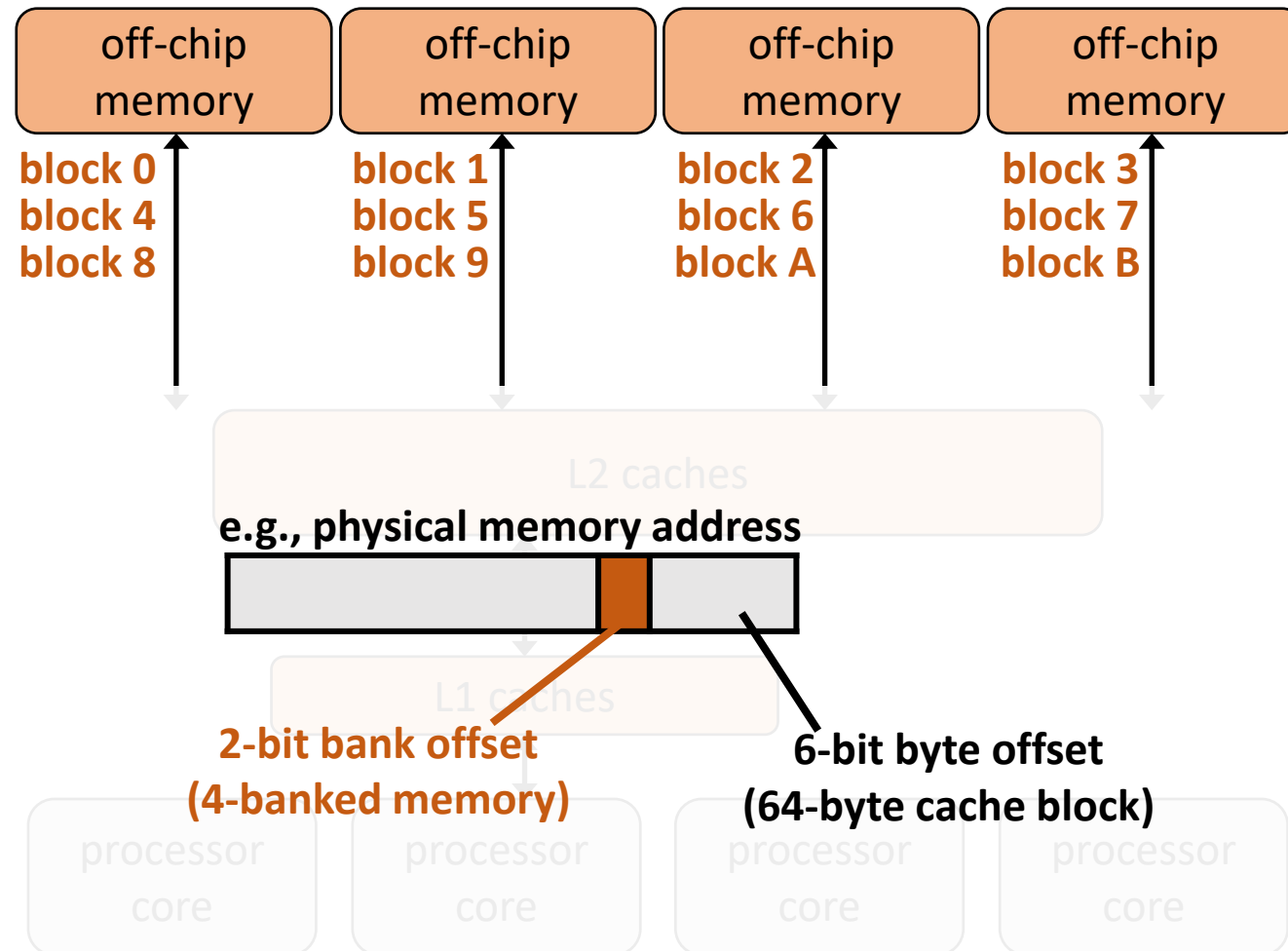
# Shared-Memory CMP – Interleaved Memory



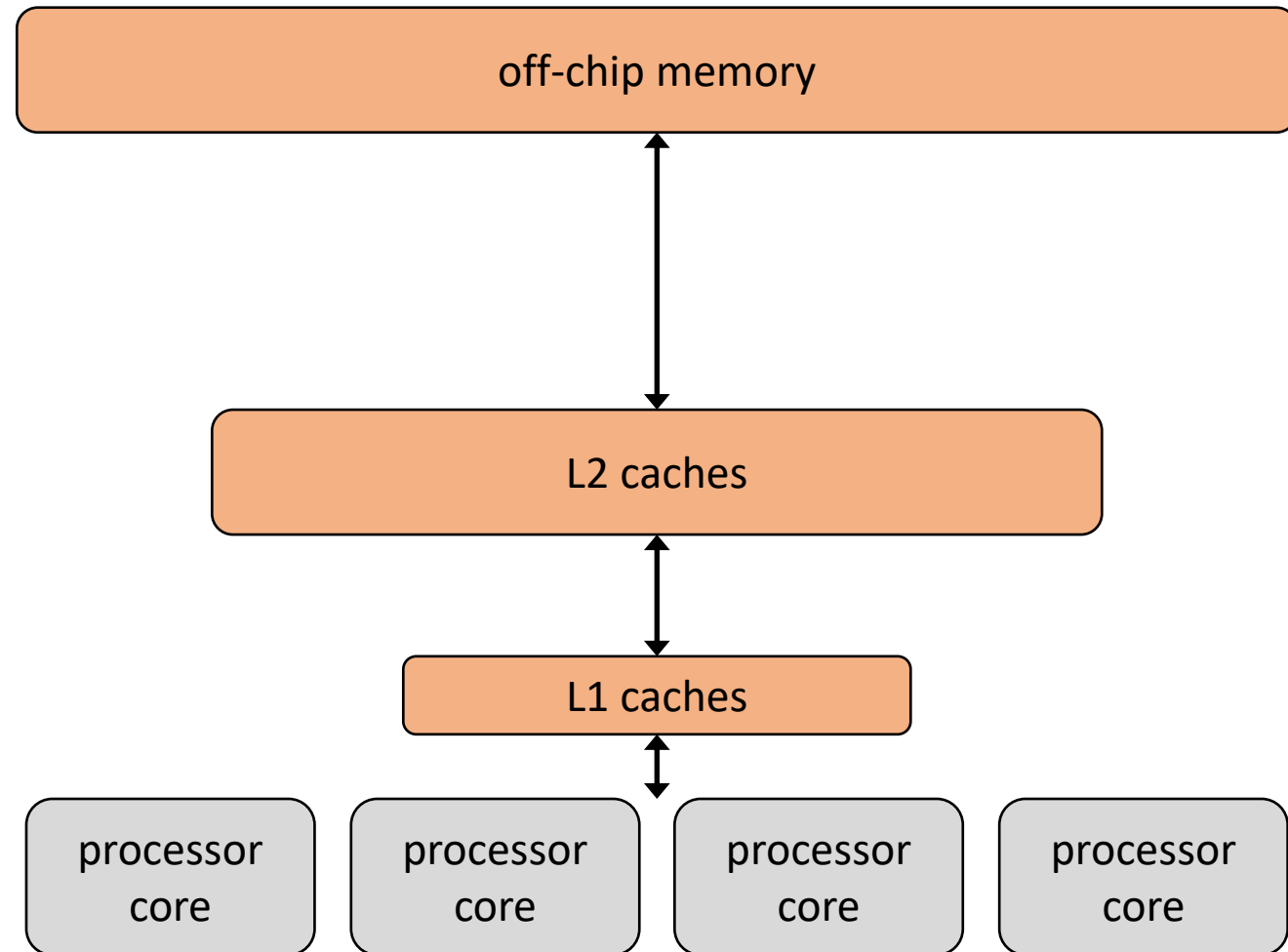
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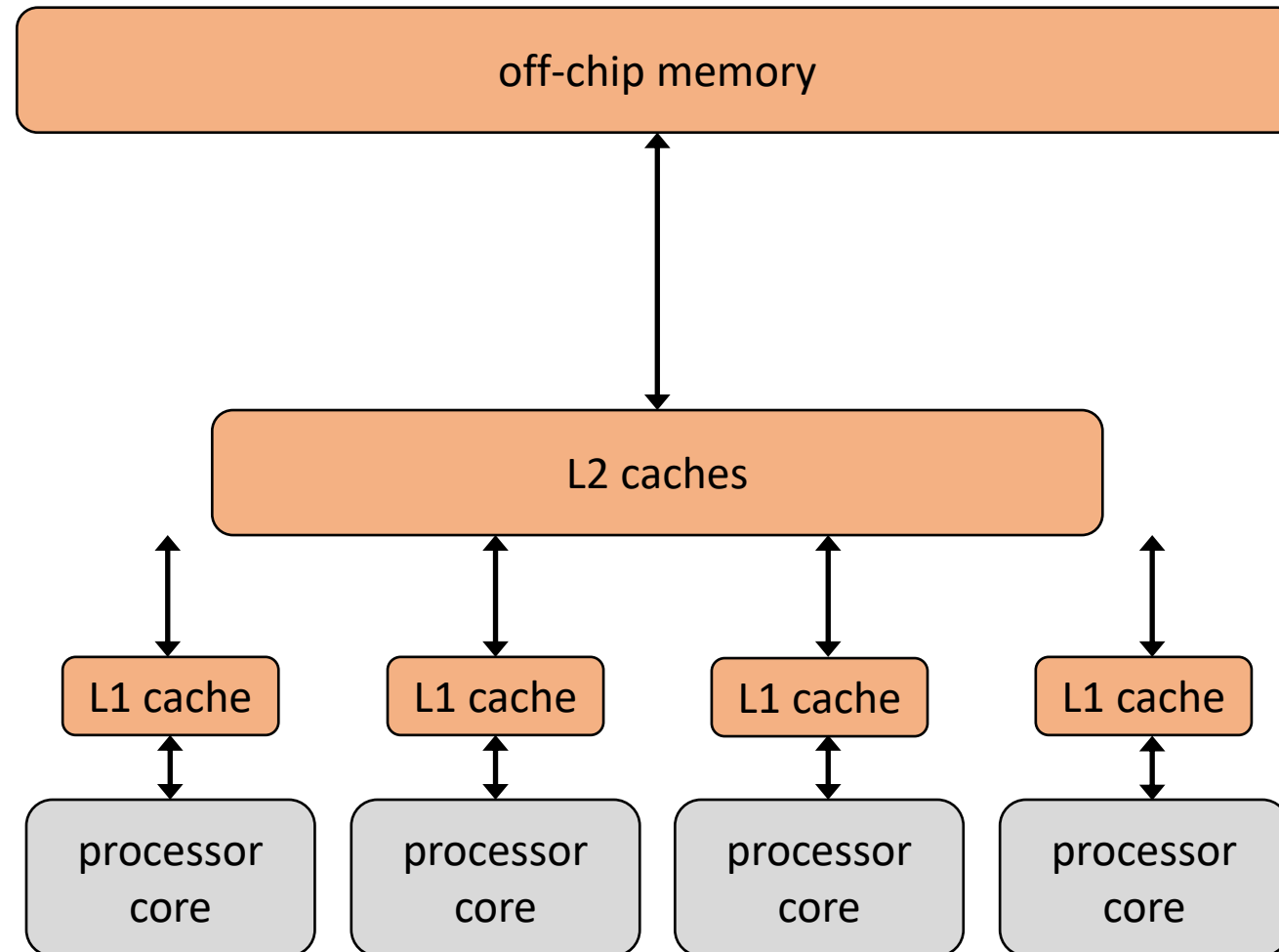
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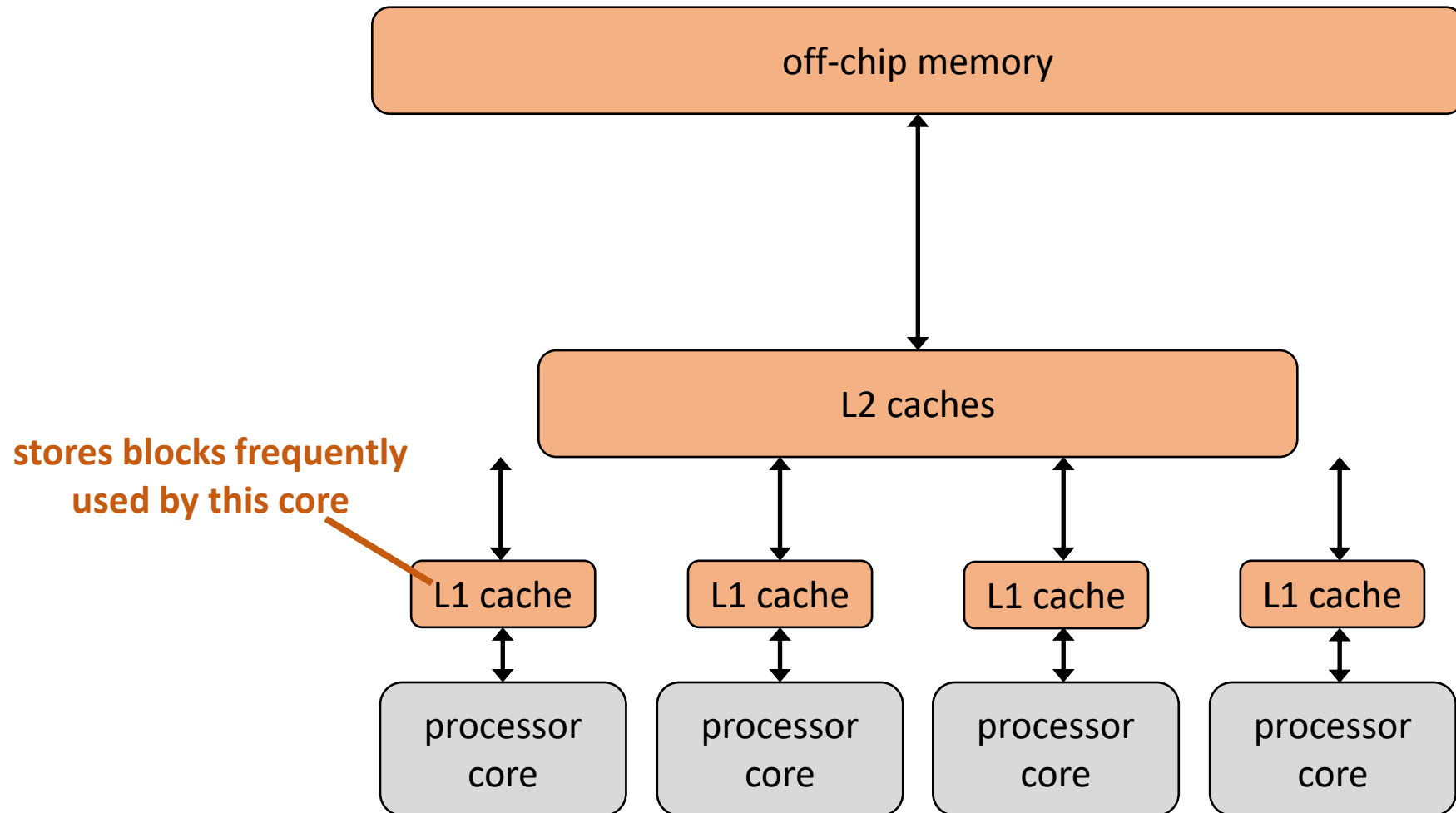
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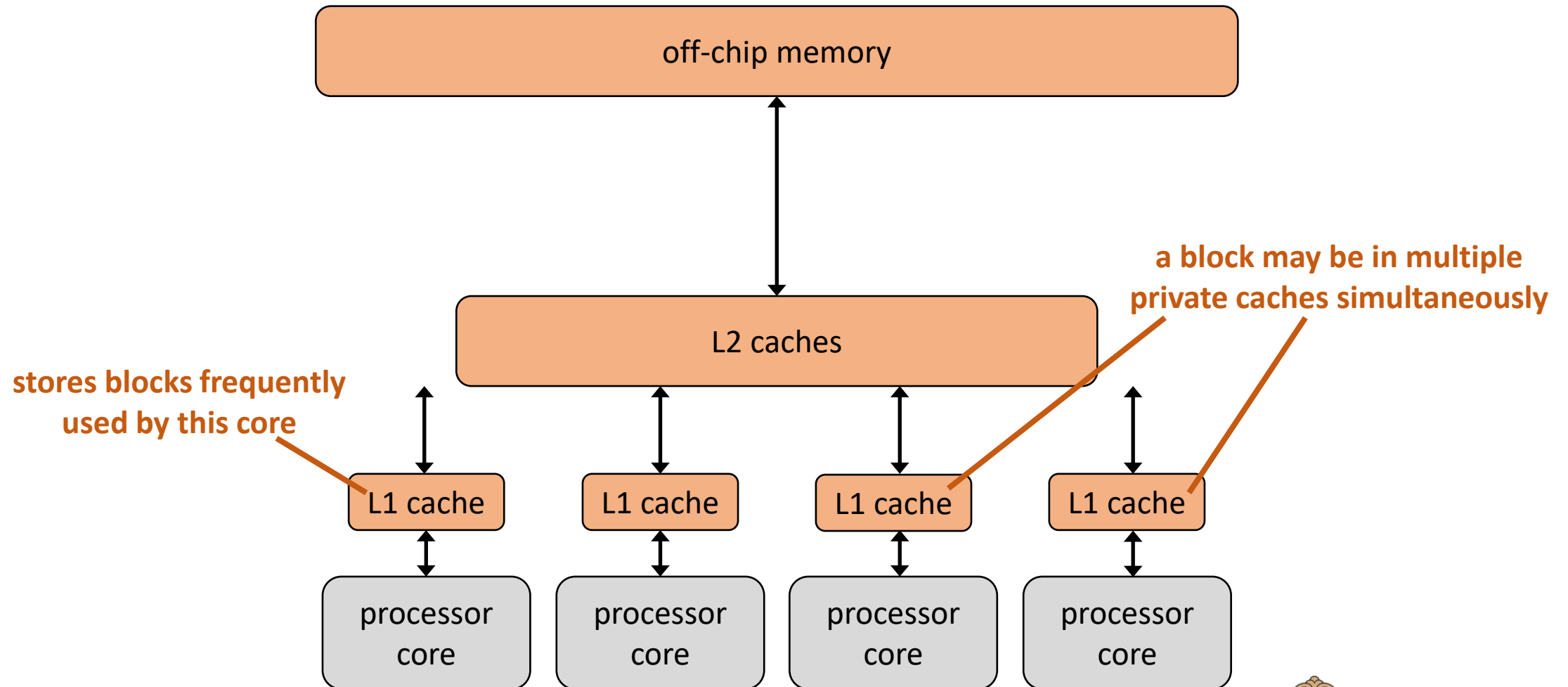
# Shared-Memory CMP – Private Caches



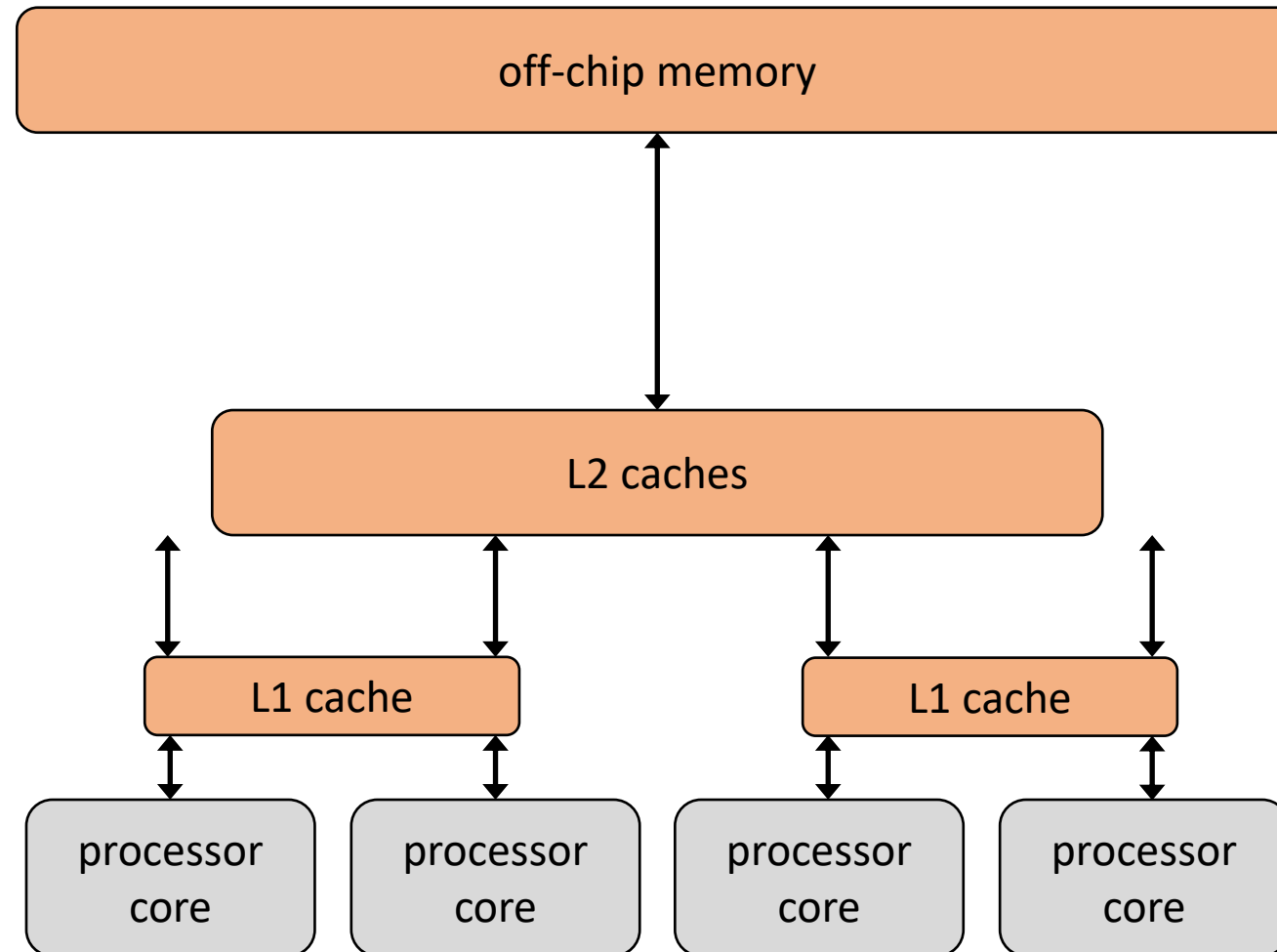
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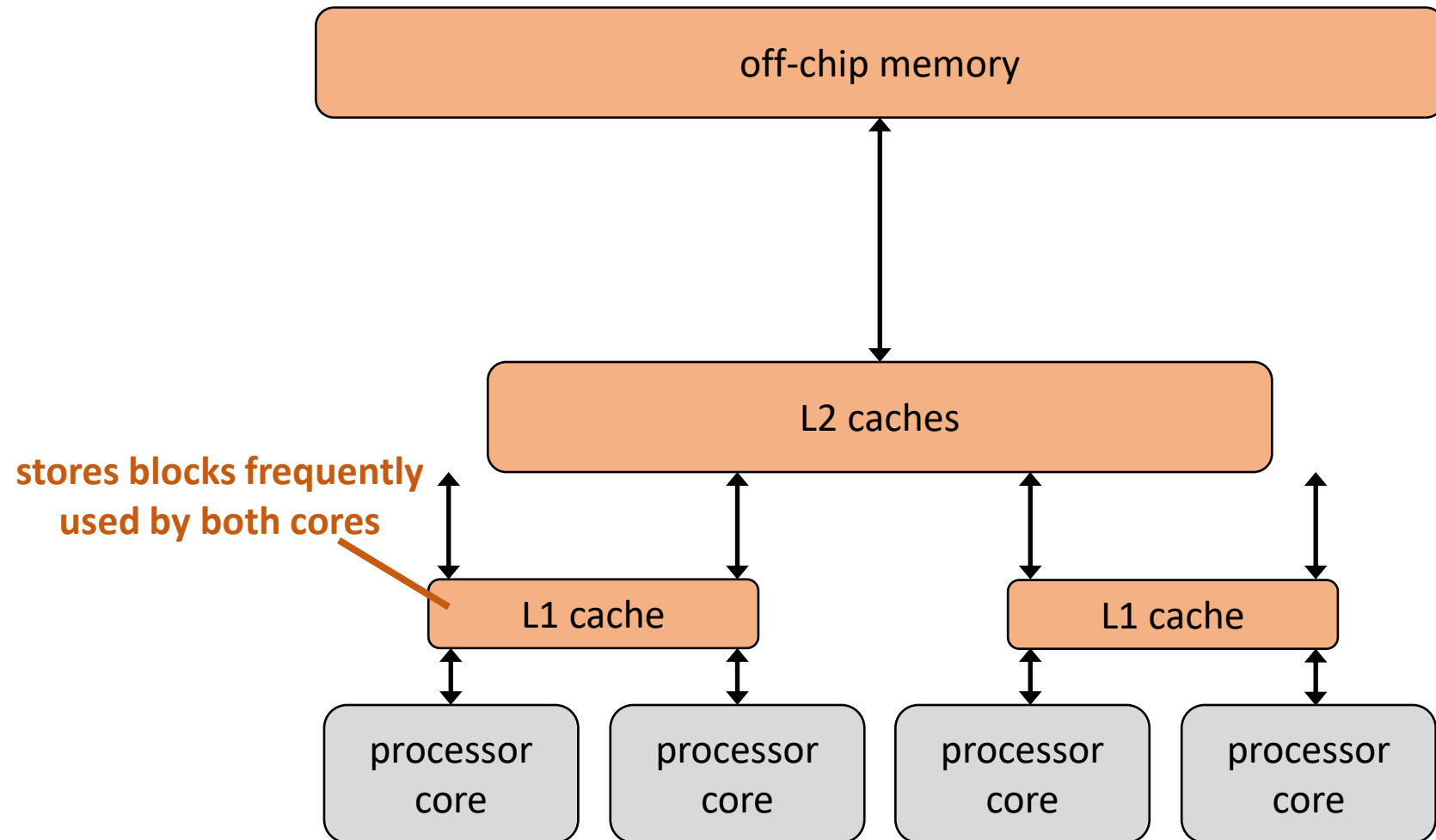
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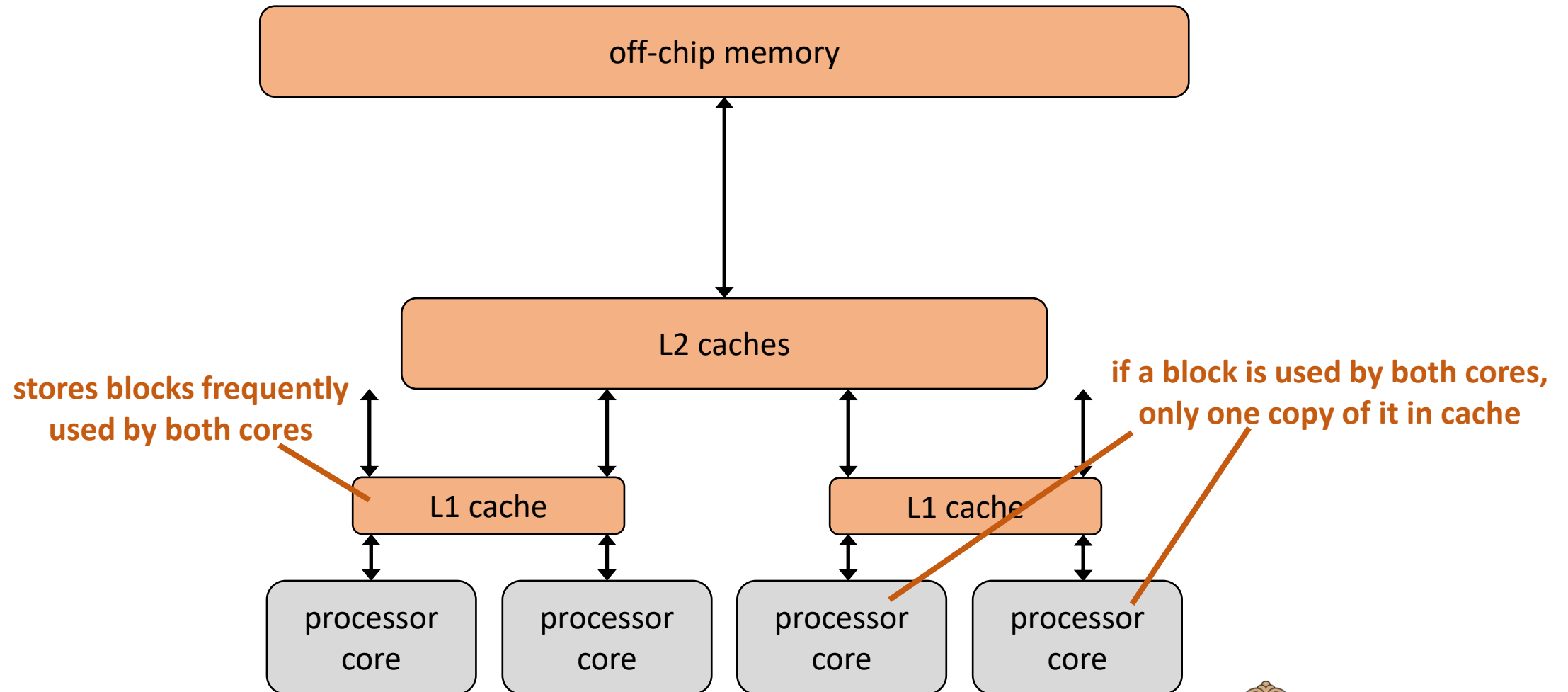
# Shared-Memory CMP – Shared Caches



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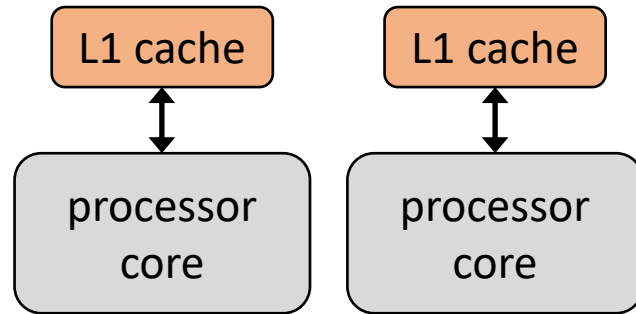


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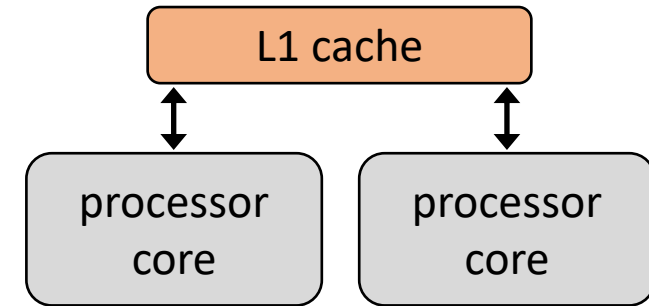
# Private and Shared Caches

## Private Caches



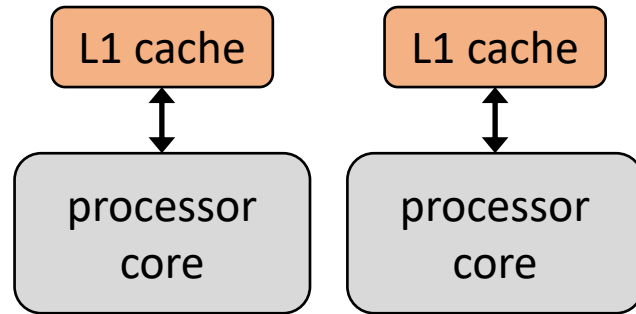
**VS**

## Shared Cache



# Private and Shared Caches

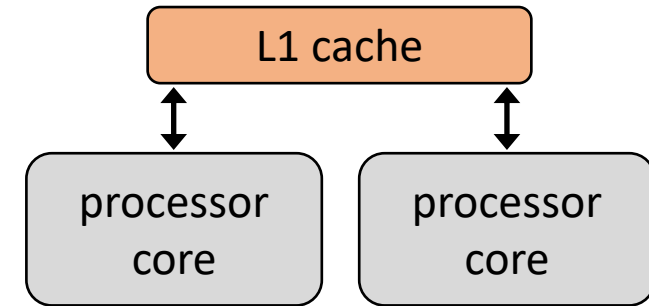
## Private Caches



- + Lower latency and energy (proportional to size of cache)
  - + Tolerant to cache contention from independent workloads
  - + More secure from other cores
- Generally used closer to cores**

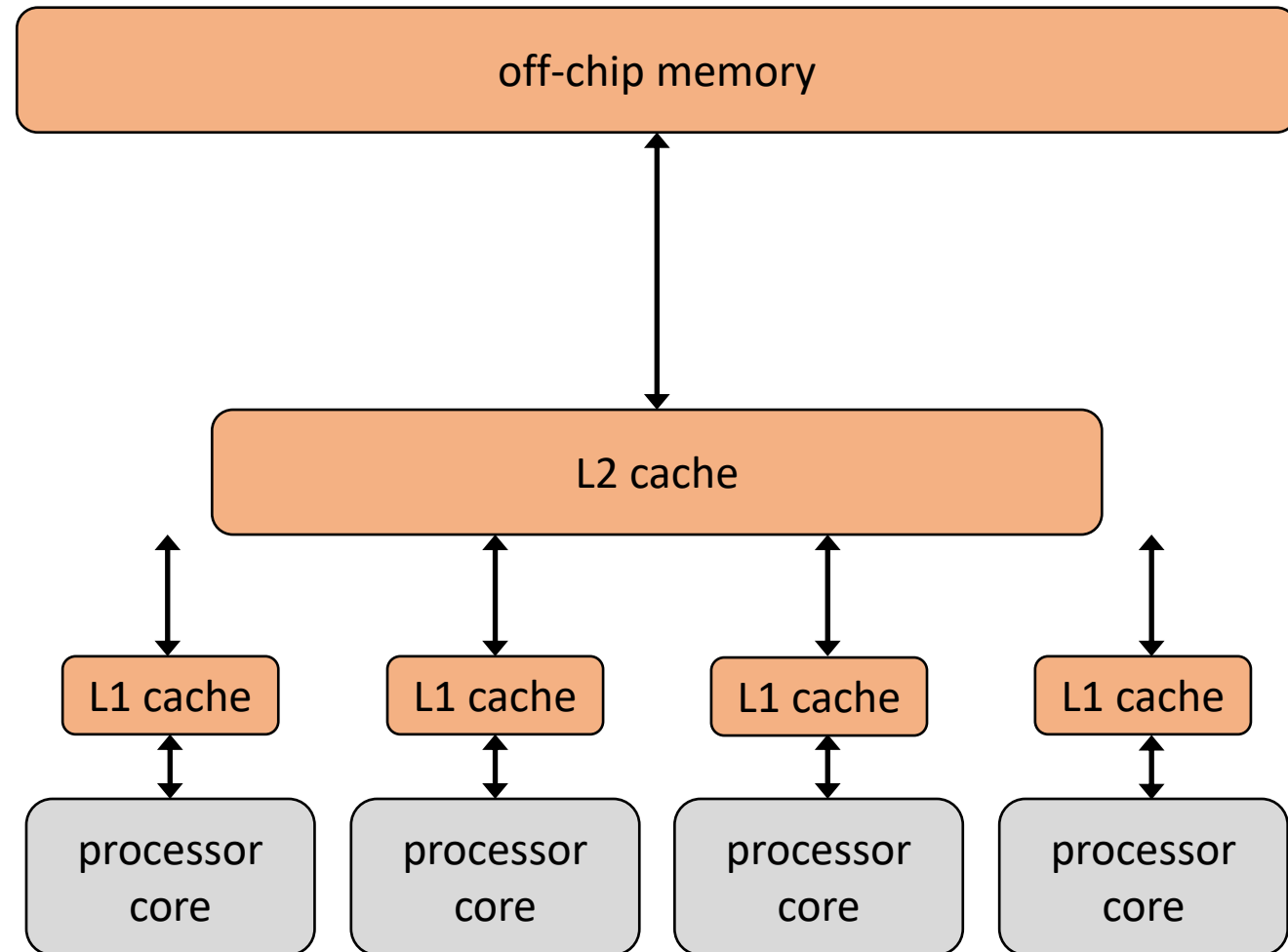
VS

## Shared Cache

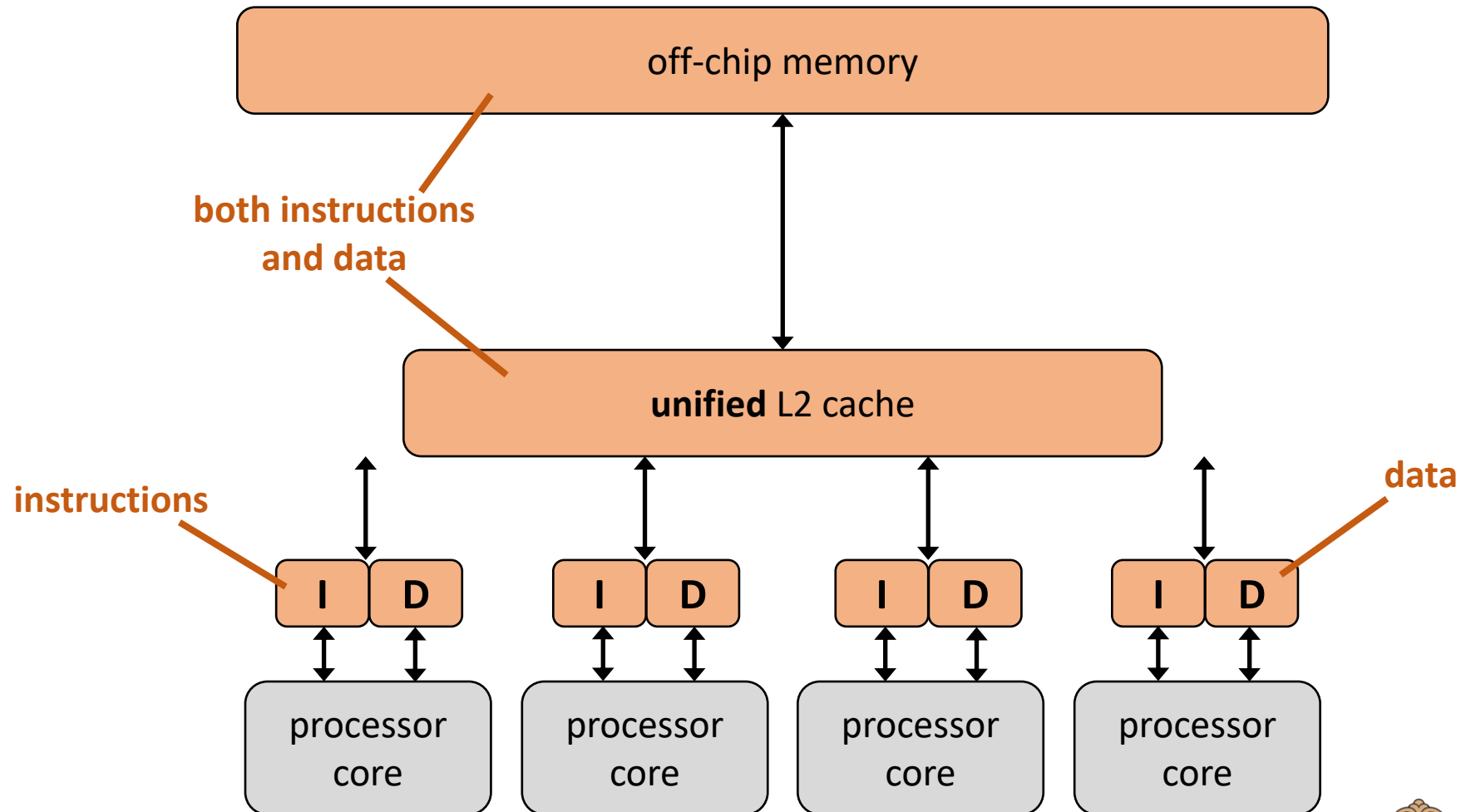


- + Higher effective capacity (no redundant blocks)
  - + More tolerant to imbalance in working set sizes
  - + Can cooperatively prefetch
- Generally used closer to memory**

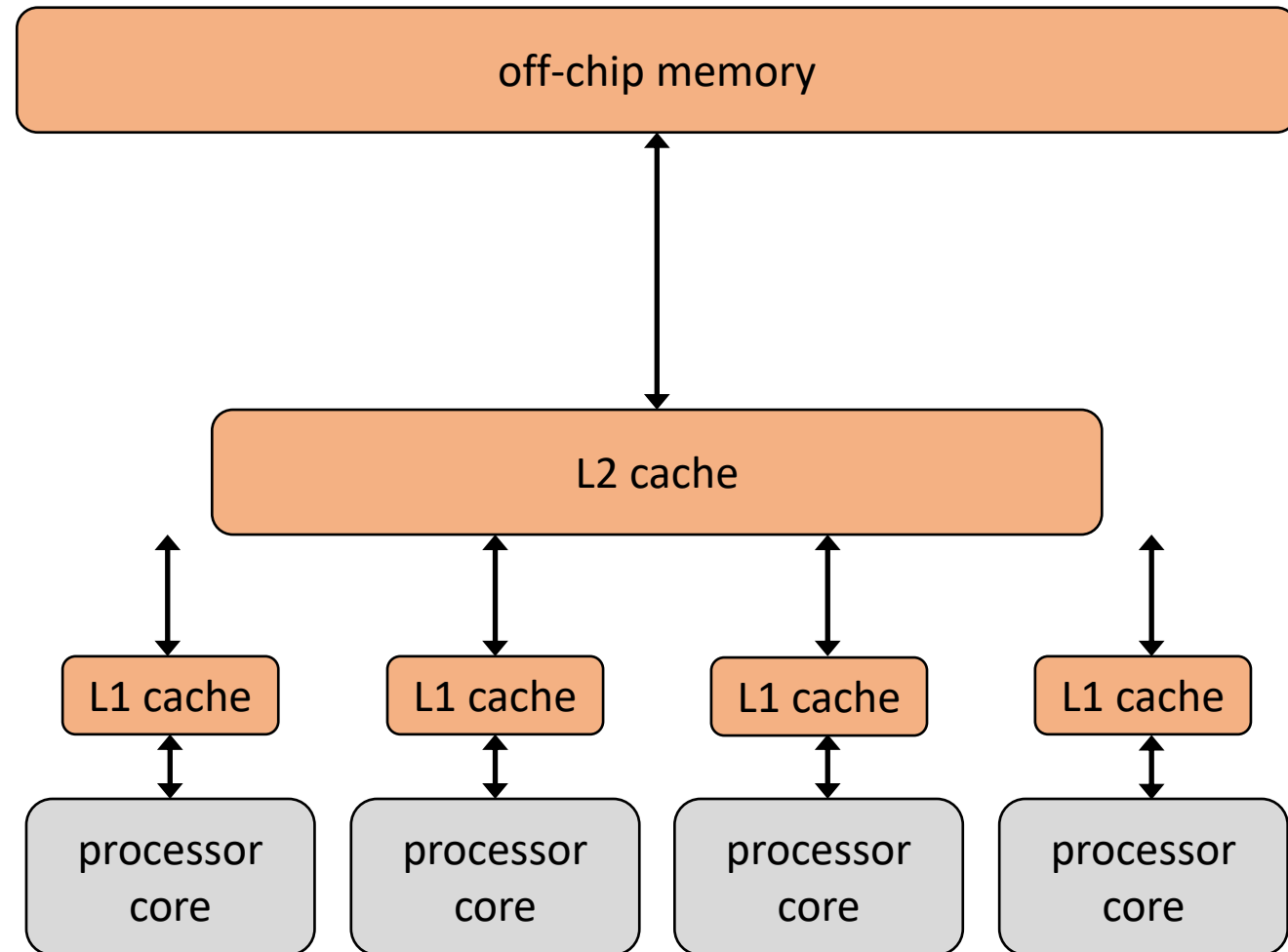
# Shared-Memory CMP



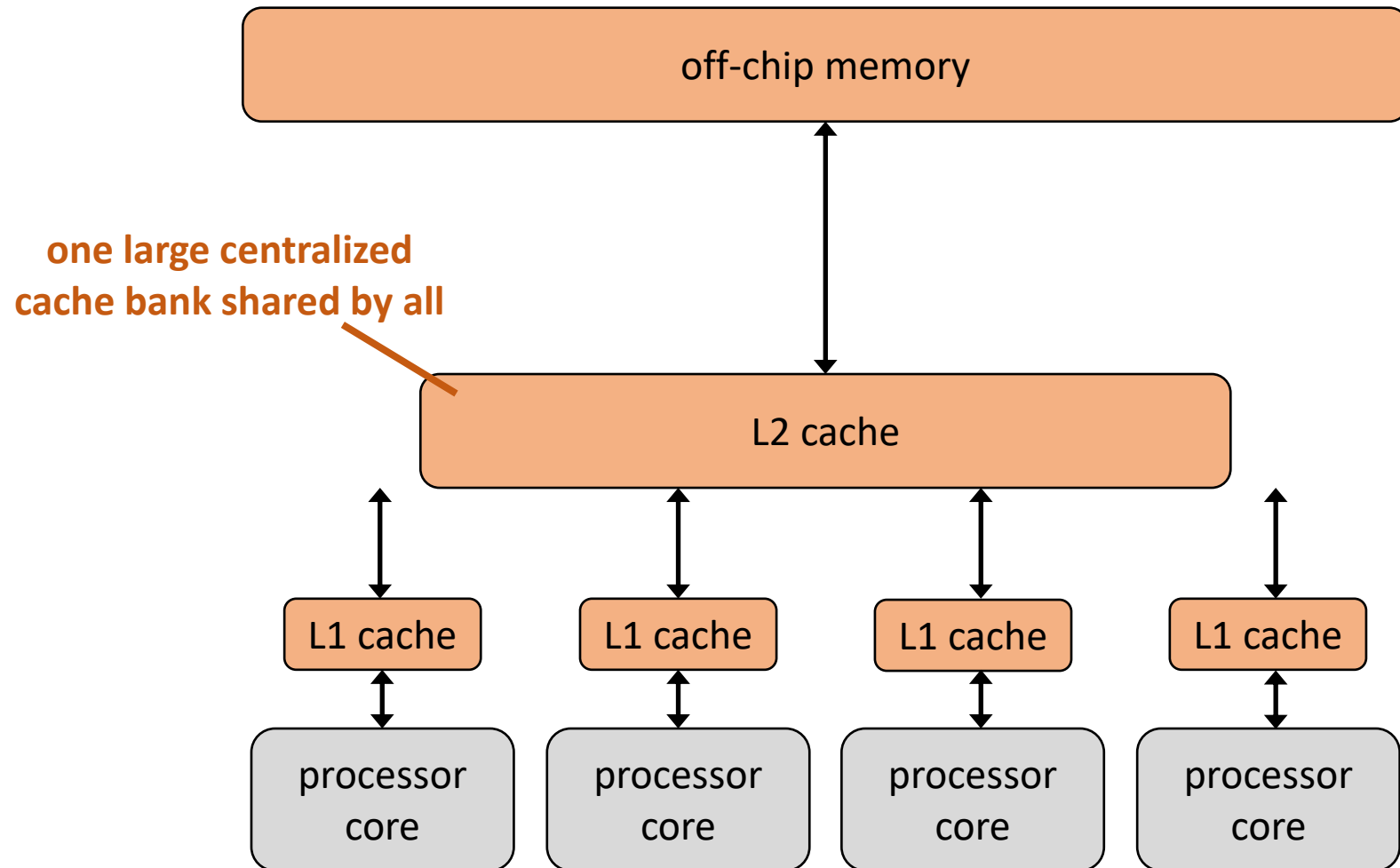
# Shared-Memory CMP – Instructions and Data



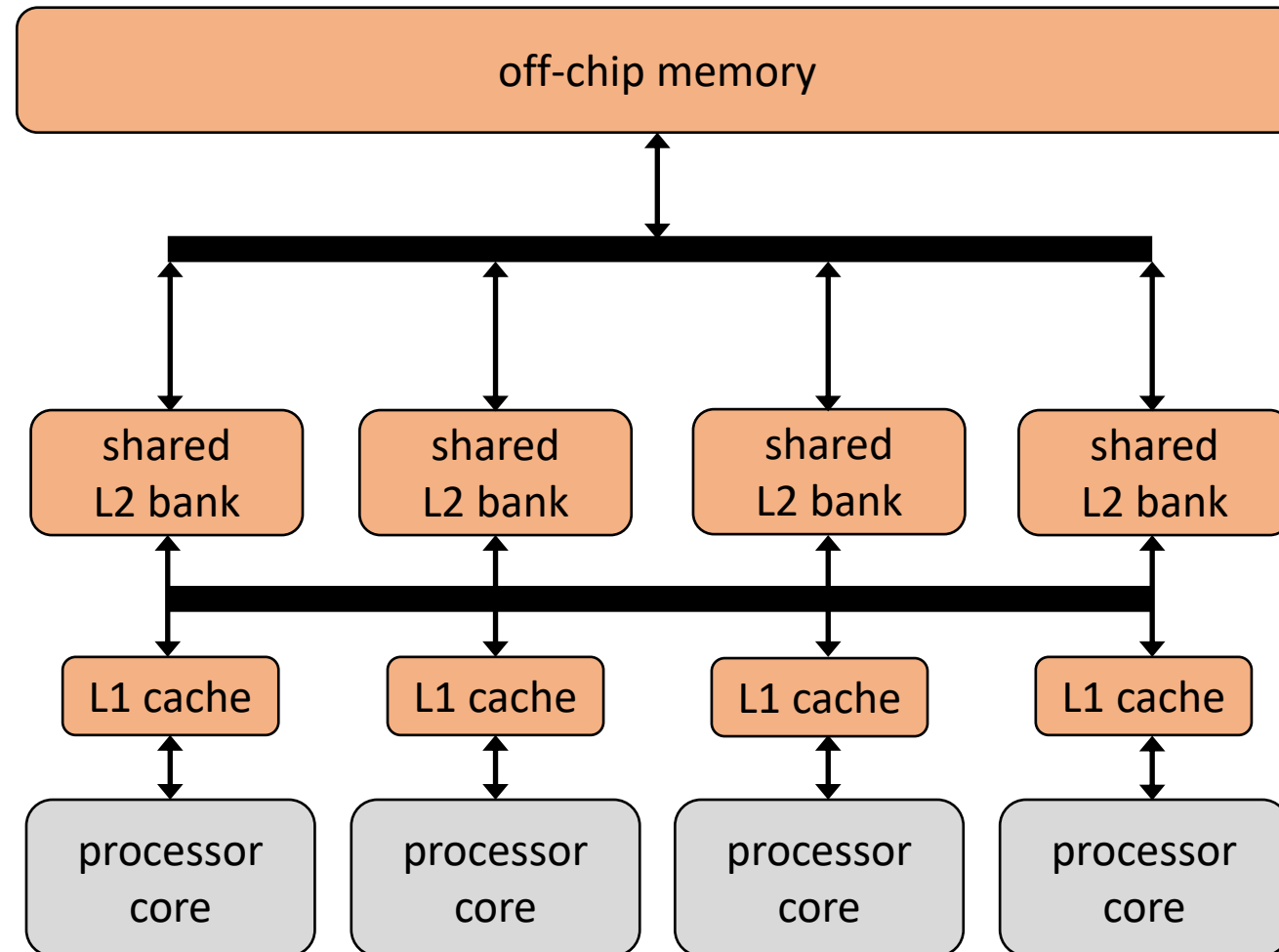
# Shared-Memory CMP



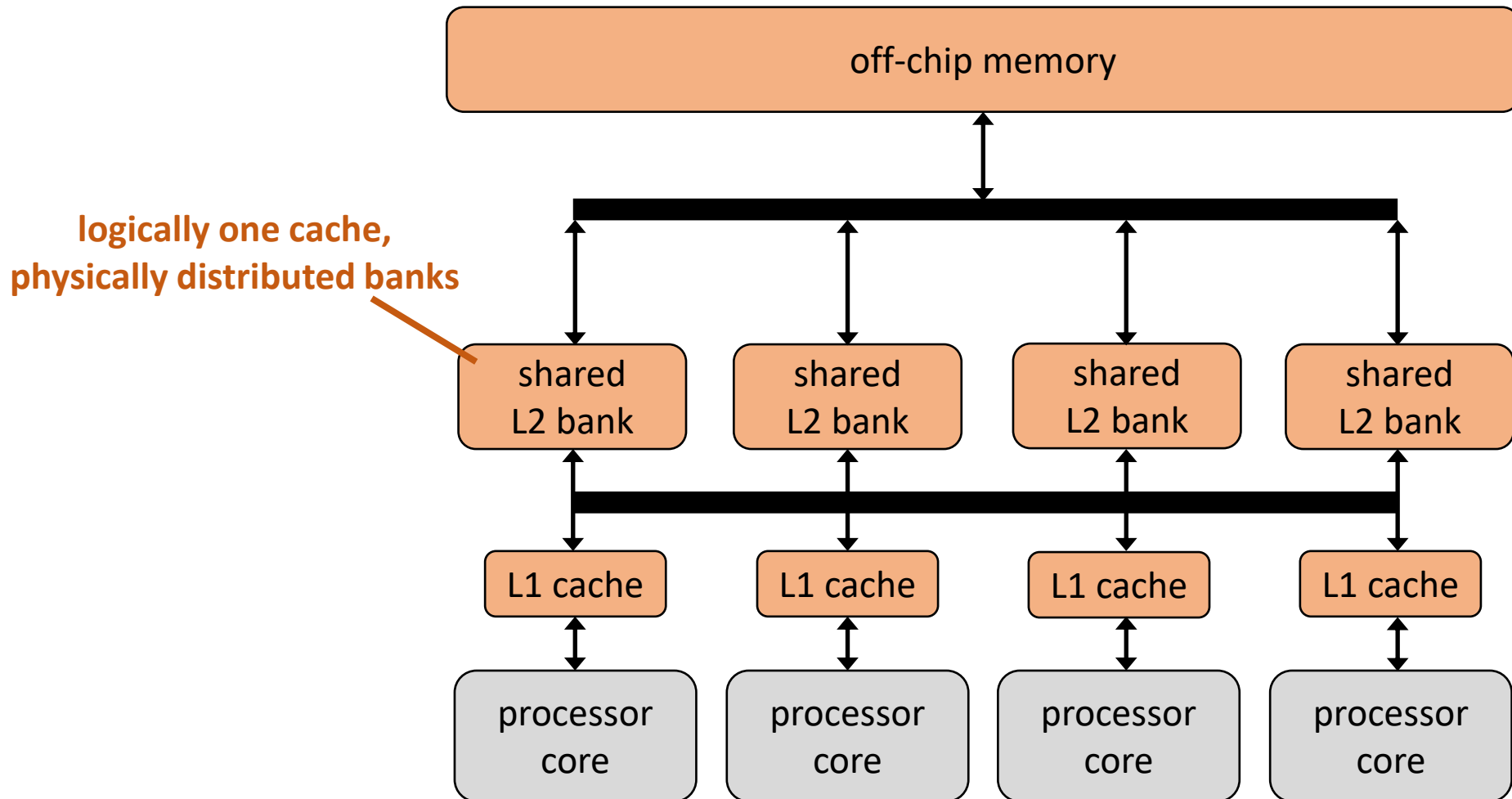
# Shared-Memory CMP – Centralized Cache



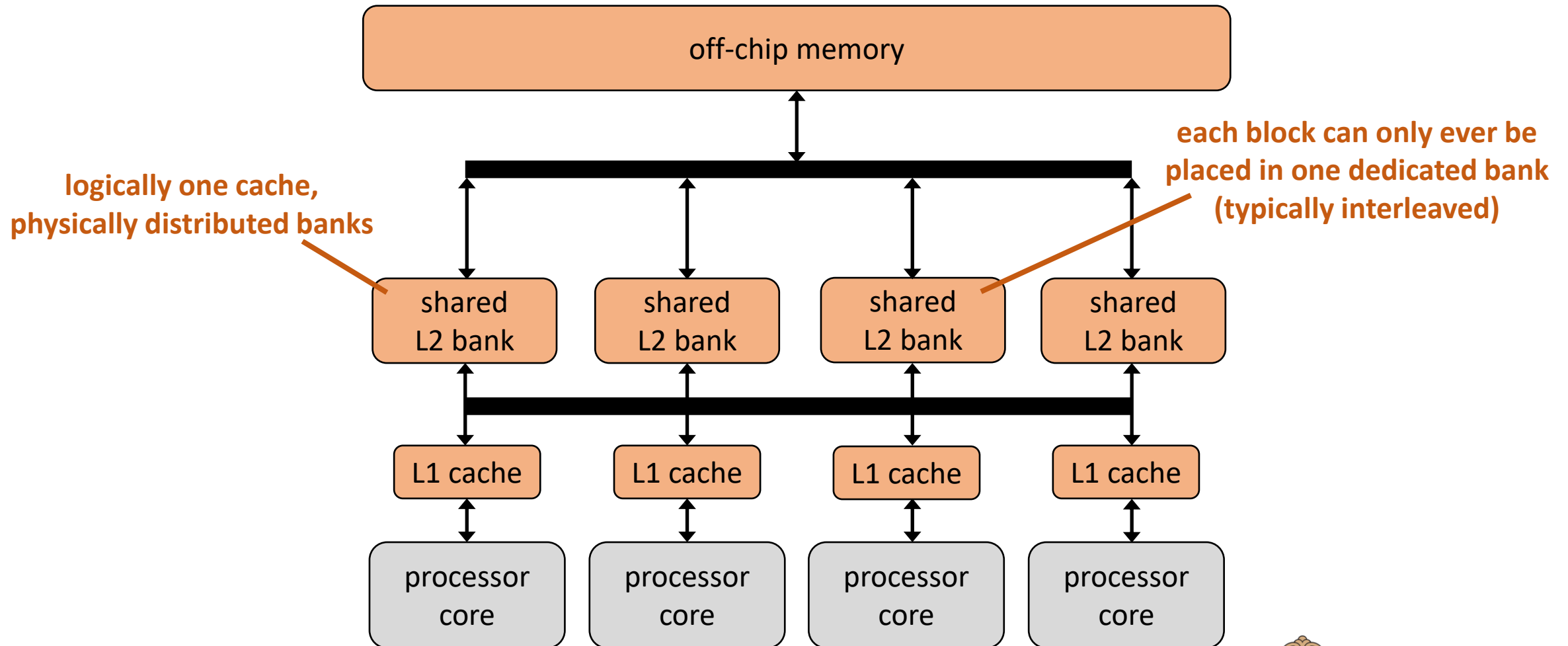
# Shared-Memory CMP – Distributed Cache



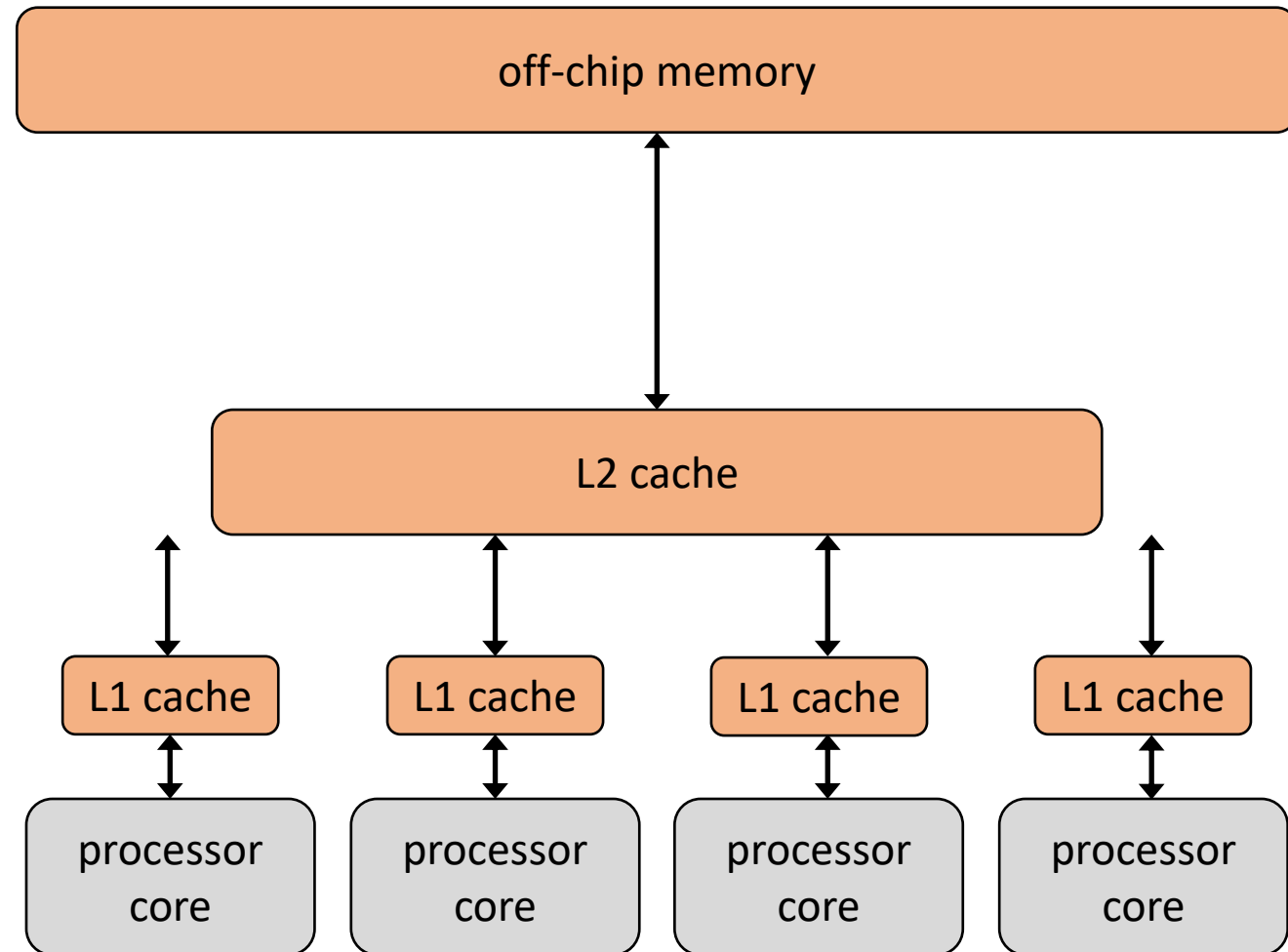
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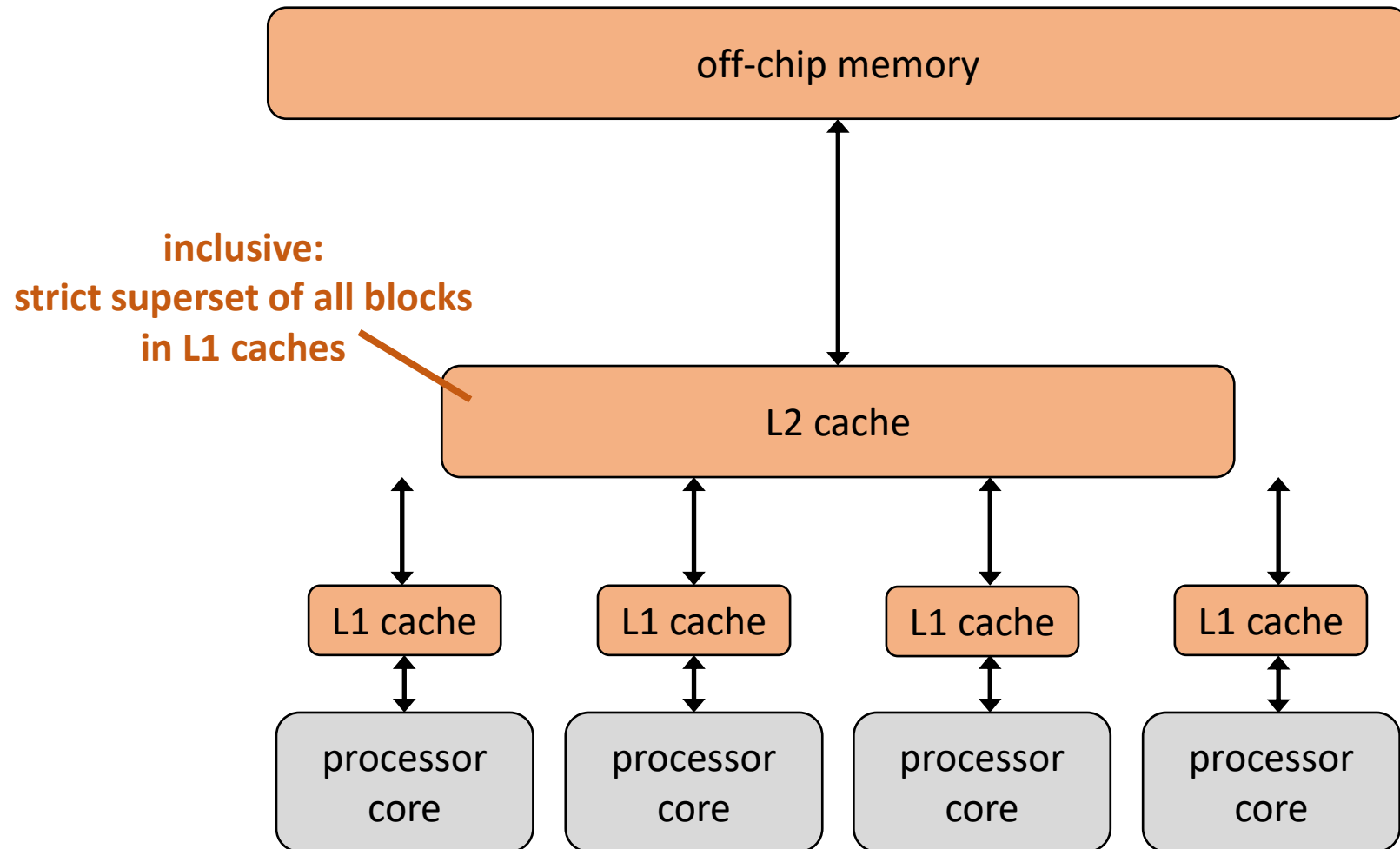
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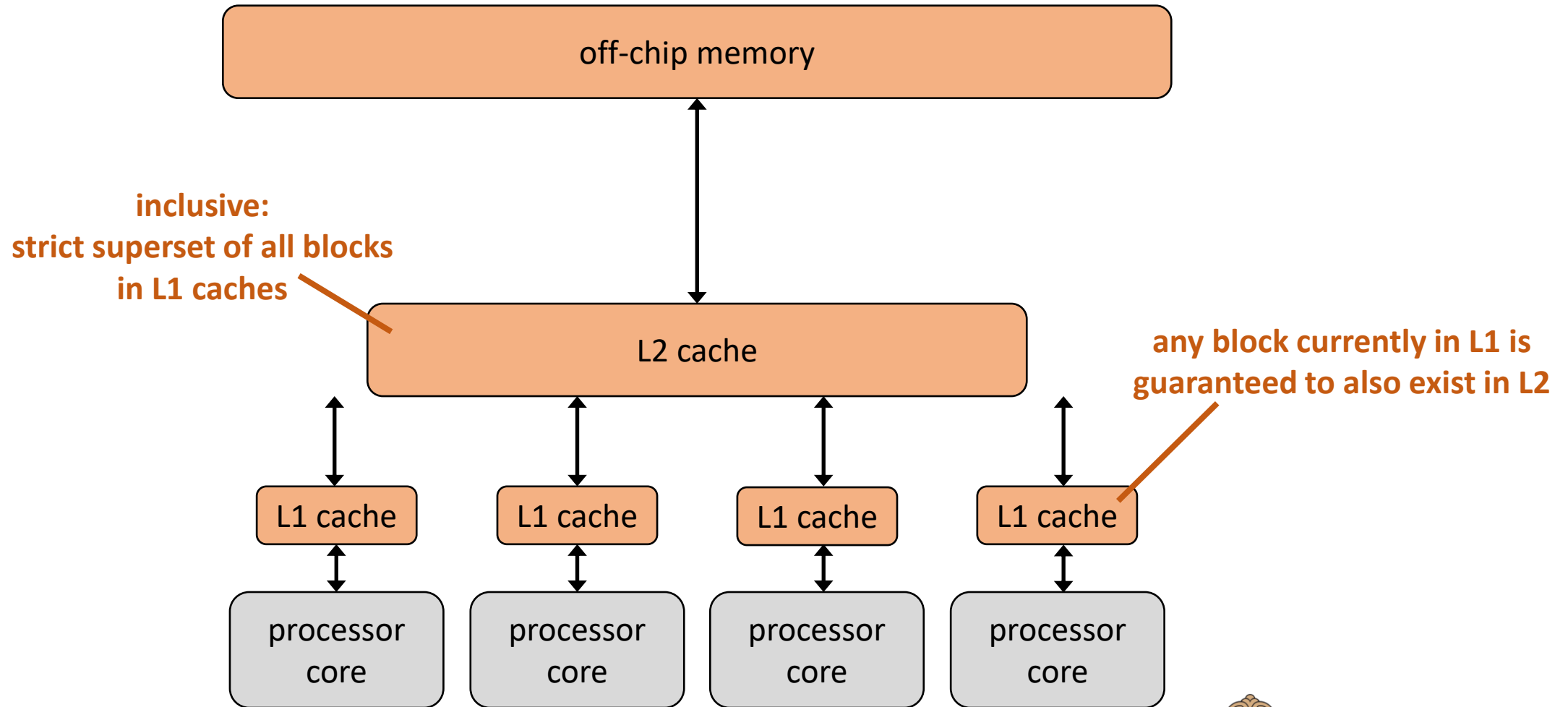
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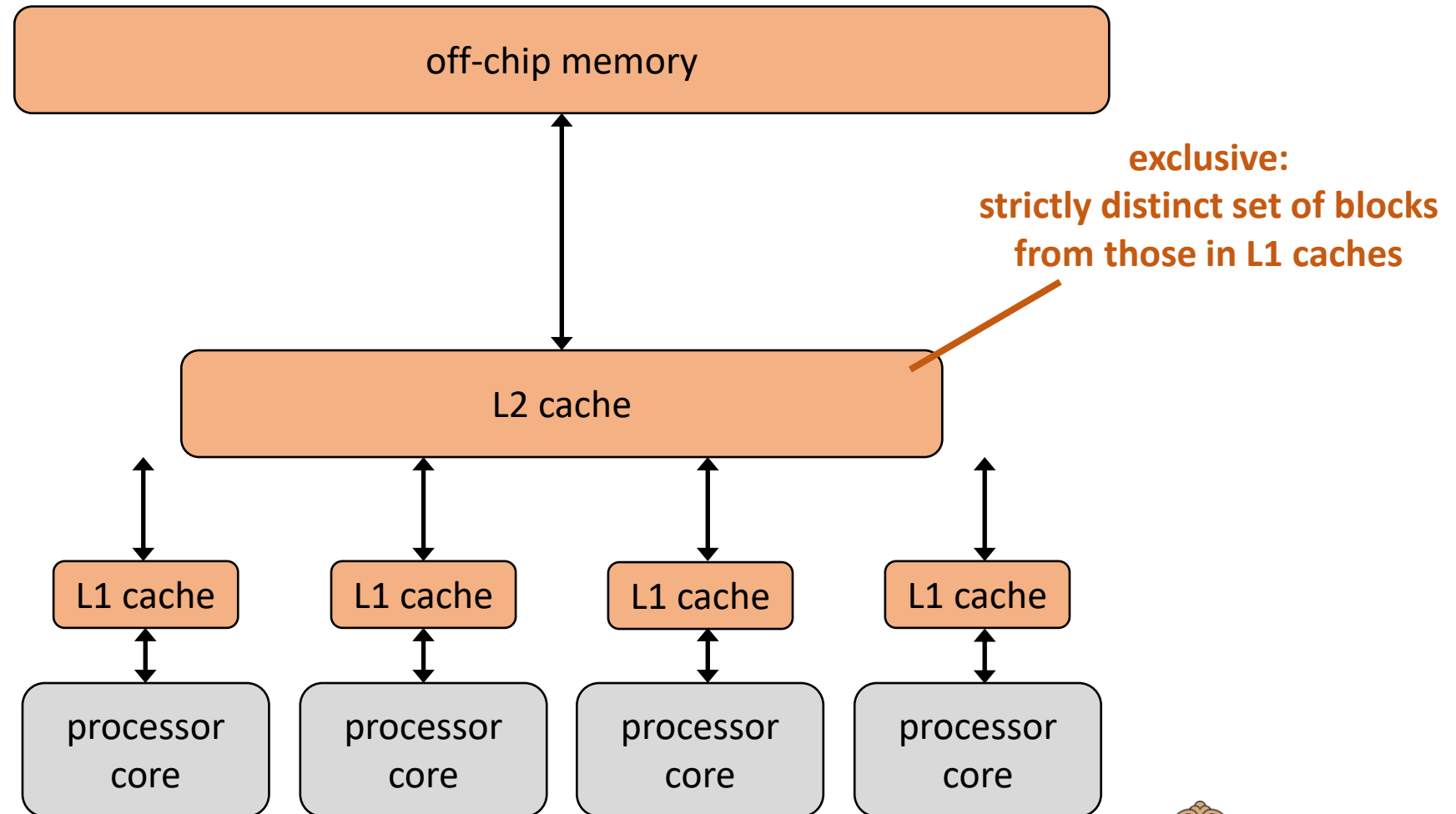
# Shared-Memory CMP – Inclusive Cache



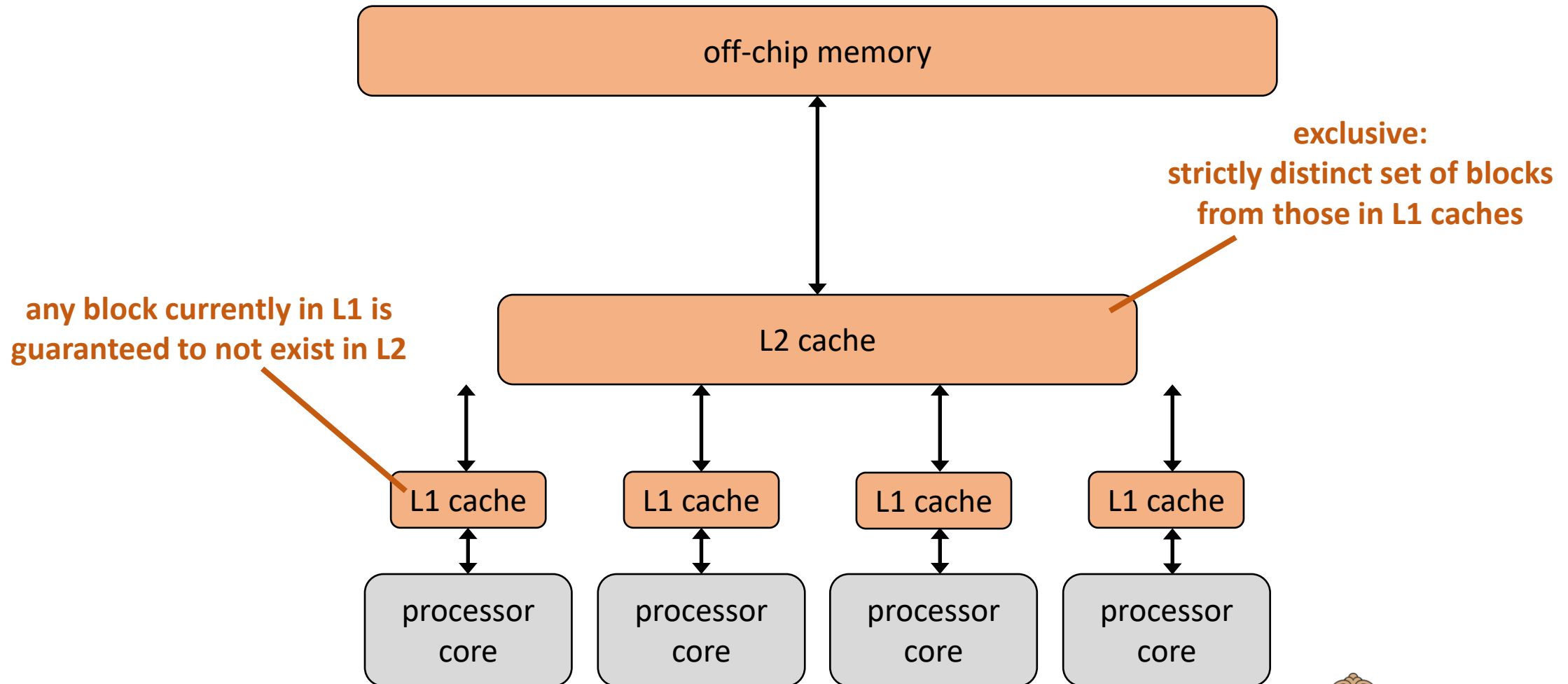
# Shared-Memory CMP – Inclusive Cache



# Shared-Memory CMP – Exclusive Cache



# Shared-Memory CMP – Exclusive Cache



# Shared-Memory CMP – Non-Inclusive/-Exclusive

